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Project Report On Thread Theory (E-commerce Website of Clothing)

by

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INDEX

Sr. No	Description	Page No.
1	Introduction	
	1.1 Project description	1
	1.2 Project Profile	3
2	Environment Description	
	2.1 Hardware and Software Requirements	4
	2.2 Technologies Used	5
3	System Analysis and Planning	
	3.1 Existing System and its Drawbacks	8
	3.2 Feasibility Study	9
	3.3 Requirement Gathering and Analysis	12
4	Proposed System	
	4.1 Scope	15
	4.2 Project Modules Functionalities	18
	4.3 Module-Wise Objectives, Functionalities, and Constraints	21
5	Detail Planning	
	5.1 Data Flow Diagram / UML	23
	5.2 Use Case Diagram	29
	5.3 Process Specification / Activity Flow Diagram	32
	5.4 Entity-Relationship Diagram / Class Diagram	35
6	System Design	
	6.1 Database Design	36
	6.2 User Interface Design	41
7	Software Testing	
	7.1 Unit Testing	70
	7.2 Integration Testing	70
	7.3 System Testing	71
8	Future Scope of Enhancements	72
9	Conclusion	73
10	Bibliography & Reference	74

Introduction

1.1 Project Description

Thread Theory is an innovative, cutting-edge e-commerce platform designed to redefine the online fashion retail experience. By bridging the gap between physical retail and digital convenience, Thread Theory offers customers a highly interactive and personalized way to shop for apparel.

The standout feature of the platform is its advanced **3D Virtual Try-On system**. Unlike traditional e-commerce stores that rely solely on static 2D images, Thread Theory leverages computer vision and 3D rendering to allow users to virtually "wear" clothing in real-time. This immersive approach boosts buyer confidence, reduces return rates, and delivers a highly engaging shopping journey tailored to the modern consumer.

1.1.1 Technology Stack

Built on a robust, modern MERN architecture combined with state-of-the-art WebGL and AI libraries, the platform ensures blazing-fast performance, high visual fidelity, and secure transactions:

- **Frontend UI: React.js** (delivering a highly dynamic and responsive component-based user interface) styled with **Tailwind CSS** (ensuring a sleek, modern, and mobile-first design system).
- **Backend API: Node.js & Express.js** (handling secure API routing, payment gateways, and business logic).
- **Database: MongoDB** (a NoSQL database managing scalable product catalogs, user profiles, and order histories).
- **3D Rendering & Augmented Reality:** * `@react-three/fiber` and `@react-three/drei` (managing the <Canvas>, 3D environment, contact shadows, and rendering .glb clothing models fetched dynamically from GitHub raw links).
- **Computer Vision (AI):** `@mediapipe/tasks-vision` (utilizing PoseLandmarker and FilesetResolver for real-time spatial body tracking and mathematical pose calculations via the user's camera).

1.1.2 Core Modules

The platform is logically divided into specialized modules to ensure secure data handling and distinct user experiences for both shoppers and store managers:

- **Customer Module:** A highly interactive, user-centric interface designed for exploring the clothing catalog and experiencing the virtual fitting room. Key features include:
 - **Immersive Virtual Try-On:** Real-time camera integration allowing users to overlay 3D .glb clothing models onto their bodies using dynamic pose estimation.
 - **Advanced Catalog Exploration:** Seamless filtering (by category, size, color, and price) with fast-loading product galleries.
 - **User Management:** Secure profiles, personalized wishlists, intuitive cart management, and streamlined checkout.
 - **Performance Optimization:** Use of `useProgress` and `<Html>` from React Three Drei to provide smooth loading states for heavy 3D assets.
- **Admin Module:** A comprehensive, secure backend dashboard built for platform administrators to oversee inventory and business operations. Key features include:
 - **Dynamic Inventory Management:** Full CRUD (Create, Read, Update, Delete) operations for clothing items, including the ability to link .glb 3D model URLs directly from GitHub.
 - **Order & User Tracking:** Tools to manage customer accounts, process orders, and track fulfillment statuses.
 - **Analytics & Business Insights:** Clear visual dashboards to track sales performance, inventory levels, and overall platform growth.

1.2 Project Profile

Project Title	Thread Theory (E-commerce Website of Clothes)
Duration	4 Months
Front-End	React , Tailwind CSS , JavaScript
Back-End	Express.js , Node.js , cors
Database	MongoDB
Other Tools	Visual Studio Code, Vite , Microsoft Office, Chrome, Postman , Canva .
Guided By	Mr. Smit Parekh
Submitted To	SDJ International College, Vesu.

Environment Description

2.1 Hardware and Software Requirements

Hardware Requirements

Client Side (Customer's Device): Because Thread Theory utilizes real-time 3D rendering and computer vision, the client hardware requires slightly more capability than a standard text-and-image e-commerce site.

- **Processor:** Modern Multi-core CPU (e.g., Intel i3/Apple Silicon or equivalent mobile processor).
- **RAM:** Minimum 4 GB (8 GB recommended for smooth 3D Canvas rendering).
- **Graphics:** WebGL-compatible integrated or dedicated GPU.
- **Peripherals:** Functioning Webcam (Strictly required for the Virtual Try-On feature).
- **Display:** Minimum 720p resolution; optimal experience on 1080p.

Development Side (Developer's Machine):

- **Processor:** Intel Core i5 / AMD Ryzen 5 / Apple M1 or above.
- **RAM:** 16 GB or above (Required for running local Node servers, MongoDB, React dev servers, and rendering 3D .glb assets simultaneously).
- **Storage:** 500 GB SSD or above for fast read/write speeds during compilation.
- **Graphics:** Dedicated GPU preferred for testing Three.js/WebGL performance.
- **Display:** Full HD (1920x1080) or higher.

Server Side (Hosting Environment):

- **Processor:** Quad-Core CPU or above.
- **RAM:** 16 GB or above.
- **Storage:** 1 TB NVMe SSD or above.
- **Network:** High-bandwidth, low-latency connection to handle rapid API responses and concurrent user sessions.

Software Requirements

Client Side (Customer's Browser):

- **Operating System:** Windows / macOS / Linux / Android / iOS.
- **Browser:** Chrome, Firefox, Safari, or Edge (Must be updated to the latest version to support WebGL and Camera API permissions).

Development Side (Developer Tools):

- **Operating System:** Windows 10/11, macOS, or Linux.
- **Front-End:** React.js, Tailwind CSS, @react-three/fiber, @mediapipe/tasks-vision.
- **Back-End:** Node.js, Express.js.
- **IDE/Editor:** Visual Studio Code.
- **Database Tooling:** MongoDB Compass, MongoDB Atlas.
- **API Testing:** Postman.

Server Side (Hosting):

- **Web Server Engine:** Node.js Runtime Environment.
- **Database:** MongoDB Atlas (Cloud-hosted).

2.2 Technologies Used

Core Web Stack (MERN):

- **Front-End:** React.js, Tailwind CSS
- **Back-End:** Node.js, Express.js
- **Database:** MongoDB

3D & AI Engine (Virtual Try-On):

- **3D Rendering:** @react-three/fiber, @react-three/drei
- **Computer Vision:** @mediapipe/tasks-vision
- **Asset Storage:** GitHub (Raw links for .glb 3D clothing models)

Development & Testing Utilities:

- **IDE:** Visual Studio Code
- **API Testing:** Postman
- **Security Middleware:** CORS

Technologies in Detail

1. **React.js** : React is a declarative, component-based JavaScript library utilized to build Thread Theory's dynamic user interface. Its Virtual DOM ensures ultra-fast UI updates, which is critical when managing the complex state of a shopping cart alongside a heavy 3D rendering environment. It serves as the foundation for our Single Page Application (SPA).
2. **Tailwind CSS**: A utility-first CSS framework used to rapidly design Thread Theory's sleek, fashion-forward interface. By applying pre-defined utility classes, the development team ensures a highly responsive, mobile-first design that maintains visual consistency across all devices without the bloat of traditional custom CSS files.
3. **Node.js & Express.js**: Node.js is an asynchronous, event-driven runtime that allows JavaScript to execute on the server. Express.js acts as the minimalist web framework built on top of Node. Together, they form the high-performance backend API that handles secure user authentication, manages shopping cart sessions, and serves product data to the React frontend.
4. **MongoDB & Mongoose**: MongoDB is a highly scalable NoSQL database utilizing a flexible, JSON-like document structure. This schema-less design is ideal for Thread Theory's complex product catalog, allowing dynamic storage of clothing attributes (sizes, colors, stock levels) and the direct URLs to the GitHub-hosted .glb 3D models.
5. **@react-three/fiber & @react-three/drei**: This is the core of the 3D visualization engine. react-three-fiber is a React wrapper for Three.js, allowing developers to build 3D scenes using reusable React components (like the <Canvas>). The @react-three/drei helper library provides vital pre-built utilities such as <Environment> for realistic lighting, <ContactShadows> for grounding the clothing models visually, and <useProgress> coupled with <Html> to display clean loading screens while the heavy .glb files are fetched from GitHub raw links.

6. **@mediapipe/tasks-vision:** Google's MediaPipe is the artificial intelligence backbone of the Virtual Try-On feature. Utilizing the FilesetResolver to load the machine learning models and the PoseLandmarker to analyze the user's webcam feed, this technology calculates exact skeletal coordinates (landmarks) in real-time. The React frontend uses these mathematical coordinates to dynamically scale, rotate, and map the 3D clothing models onto the user's body on the screen.
7. **CORS (Cross-Origin Resource Sharing):** CORS is an essential security middleware implemented in the Express backend. It securely governs how the React frontend (running on a client browser) interacts with the Node.js API, ensuring data integrity and protecting the platform against unauthorized cross-site requests.

❖ Platform Advantages Summary

- **Immersive Customer Experience:** The integration of WebGL (Three.js) and AI (MediaPipe) transforms a standard e-commerce site into an interactive fitting room, significantly boosting buyer confidence and reducing apparel return rates.
- **Full-Stack JavaScript Efficiency:** Utilizing the MERN stack (MongoDB, Express, React, Node) alongside Three.js means the entire platform is written in JavaScript/TypeScript. This creates a highly unified, efficient development pipeline from backend database queries to frontend 3D rendering.
- **Dynamic Asset Delivery:** By utilizing raw GitHub links for .glb models instead of storing them locally, the platform maintains a lightweight server footprint while dynamically fetching heavy 3D assets only when the user requests a virtual try-on.

System Analysis and Planning

3.1 Existing System and its Drawbacks

Currently, the online apparel and fashion industry relies heavily on standard e-commerce templates and static media. While convenient, this traditional model presents significant limitations that hinder customer satisfaction and business profitability:

- **High Return Rates & Fit Uncertainty:** The biggest drawback of current clothing websites is the reliance on 2D images and generic sizing charts. Customers cannot visualize how a garment will actually look or fit on their body, leading to hesitation at checkout and massively high return rates.
- **Lack of Immersive Engagement:** Traditional platforms fail to replicate the tactile, interactive experience of a physical fitting room. The shopping journey feels flat and transactional rather than engaging.
- **Performance Bottlenecks with Heavy Media:** Legacy systems often struggle to load high-resolution image galleries quickly, leading to clunky UX and poor mobile optimization.
- **Geographical & Time Limits:** Physical clothing boutiques provide the try-on experience but restrict sales to local areas and specific operating hours, severely limiting the brand's market reach.
- **Rigid Asset Management:** Storing hundreds of high-quality images directly on the server rapidly consumes storage space and bandwidth, increasing operational costs.

Conclusion: There is a clear need for an interactive, high-performance platform like **Thread Theory**. By utilizing a modern React SPA architecture and integrating a real-time 3D Virtual Try-On system, Thread Theory bridges the gap between the confidence of physical retail and the scale of modern digital commerce.

3.2 Feasibility Study

This study confirms that the Thread Theory platform—including its advanced AI and 3D rendering capabilities—is technically possible, economically justifiable, and operationally sound before committing extensive development resources.

1. Technical Feasibility:

- **Proven Core Stack:** Utilizes the industry-standard MERN stack (MongoDB, Express.js, React.js, Node.js) combined with Tailwind CSS for rapid, scalable development.
- **Client-Side AI Processing:** By leveraging `@mediapipe/tasks-vision` (specifically `PoseLandmarker`), the heavy mathematical calculations for body tracking are processed directly in the user's browser, preventing server overload.
- **Efficient 3D Rendering:** The use of `@react-three/fiber` and `@react-three/drei` makes handling complex WebGL scenes manageable within React. Crucially, fetching .glb 3D clothing models via **GitHub raw links** ensures the core server remains lightweight and fast.
- **Graceful Degradation:** Tools like `useProgress` and `<Html>` from `Drei` ensure that while 3D assets load, the user sees a smooth, professional loading interface rather than a frozen screen.

2. Economic Feasibility:

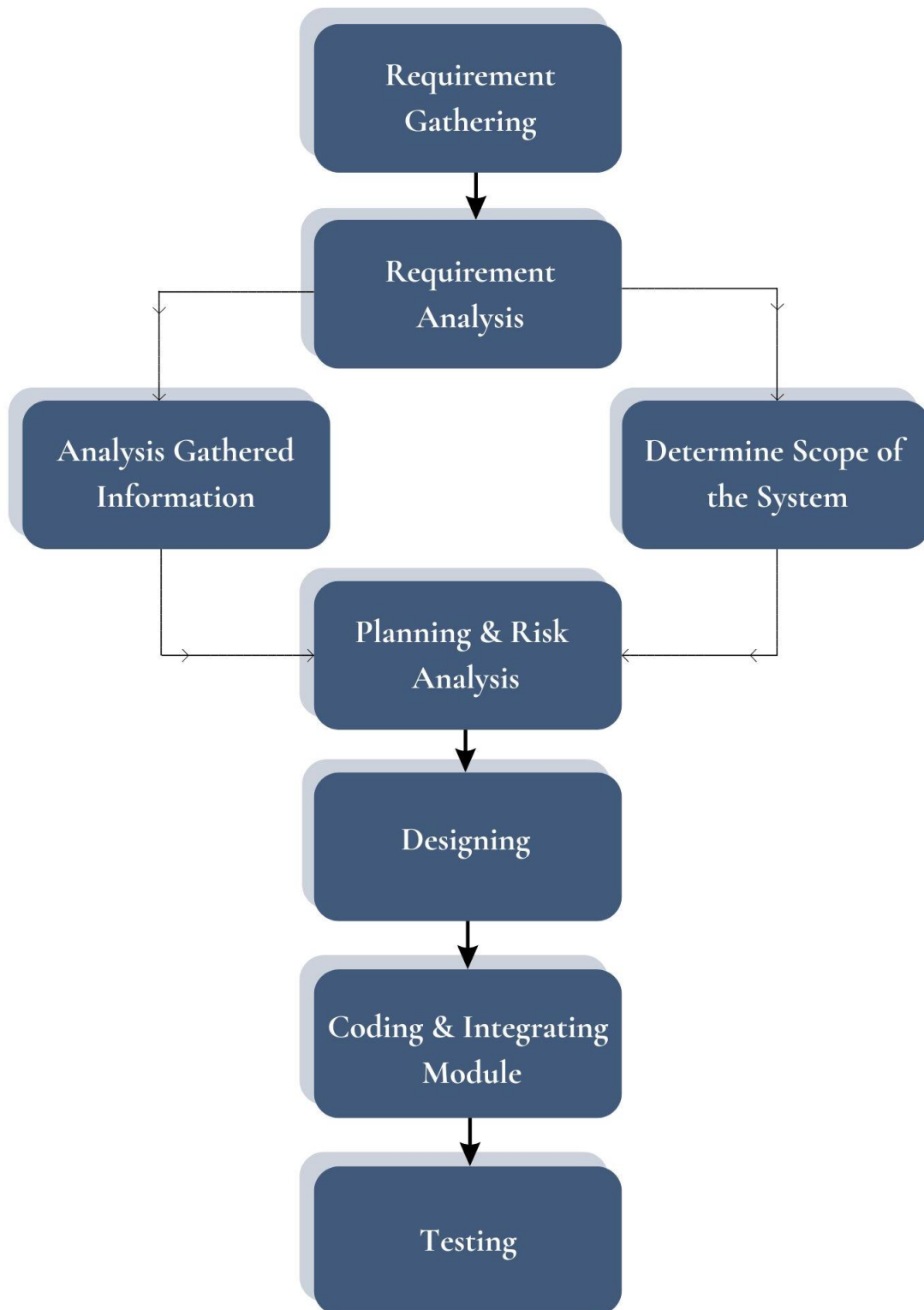
- **Zero Licensing Fees:** The entire technology stack—from React and Node.js to Three.js and Google's MediaPipe—is open-source, eliminating expensive software licensing costs.
- **Reduced Bandwidth Costs:** Hosting heavy .glb files externally (via GitHub) drastically reduces the hosting and CDN bandwidth costs typically associated with 3D web applications.
- **ROI Through Reduced Returns:** The Virtual Try-On feature directly tackles the apparel industry's biggest financial drain—product returns. By allowing users to virtually test the fit, the platform minimizes return shipping costs and maximizes profit margins.

3. Operational Feasibility:

- **Seamless Asset Management:** The backend (Node/Express/MongoDB) allows administrators to simply paste a GitHub raw URL to link a new 3D model to a product, making inventory updates incredibly fast and requiring no specialized 3D knowledge from the admin staff.
- **Intuitive UI/UX:** Tailwind CSS ensures a premium, mobile-first design. The complex <Canvas> and ContactShadows elements are abstracted away from the user, presenting them with a simple "Enable Camera to Try On" button.
- **Scalable Architecture:** MongoDB's flexible schema easily accommodates traditional product data (sizes, colors, SKUs) alongside specialized 3D configuration data.

Conclusion: The Thread Theory project is highly feasible and presents a massive competitive advantage. The open-source MERN stack mitigates financial risk, the innovative use of GitHub for .glb hosting ensures technical efficiency, and the MediaPipe integration delivers an unparalleled user experience.

❖ Task Dependency



3.3 Requirement Gathering and Analysis

This phase establishes the definitive architectural blueprint for the **Thread Theory** platform. It ensures that the chosen MERN technology stack and advanced WebGL/AI integrations accurately address modern fashion e-commerce expectations, specifically solving the industry-wide problem of high return rates through an interactive fitting experience.

1. Functional Requirements

Functional requirements define the specific behaviors, features, and interactive capabilities the system must perform. These are categorized by user access levels:

Customer Module:

- **Virtual Try-On Engine (Core Feature):** The system must request secure camera access to map the user's skeletal landmarks in real-time. Utilizing `@mediapipe/tasks-vision` (specifically `FilesetResolver` and `PoseLandmarker`), the frontend must calculate shoulder distance and torso positioning. It must then seamlessly overlay dynamic 3D clothing models (.glb format) onto the user's live video feed within a `@react-three/fiber <Canvas>`.
- **Product Discovery & Interaction:** The interface must provide advanced search filters (by category, size, color, and price), high-resolution 2D image galleries, and a highly responsive shopping cart managed via React state.
- **Account Management:** Secure user registration, personalized profiles, wishlists, real-time order tracking, and dynamic sizing preferences stored in the MongoDB database.

Admin & Staff Module:

- **3D Asset & Catalog Management:** Administrators must have full CRUD (Create, Read, Update, Delete) capabilities for the master clothing catalog. Crucially, the admin dashboard must allow staff to input and link external **GitHub raw URLs** for the .glb 3D assets associated with each garment, preventing server bloat.
- **Order & Inventory Fulfillment:** Tools for processing incoming apparel orders, updating shipping statuses, and automatically monitoring dynamic inventory levels across different sizes and variants to prevent overselling.
- **Business Oversight:** Complete user management controls and access to comprehensive sales analytics, revenue reports, and platform performance metrics.

2. Non-Functional Requirements

Non-functional requirements define the quality, performance, hardware interaction, and security standards of the platform:

Performance & Loading Optimization:

- **Asynchronous 3D Asset Management:** Because .glb files hosted on GitHub can be large, the system must guarantee that the browser does not freeze during downloads. The platform must utilize @react-three/drei's useProgress hook paired with the <Html> component to display a smooth, non-blocking HTML loading interface *inside* the 3D canvas until the asset is fully parsed and ready to render.
- **Fast Page Rendering:** The core website must guarantee fast initial load times powered by React's efficient Virtual DOM rendering and Express.js's asynchronous API routing.

Security & Privacy Standards:

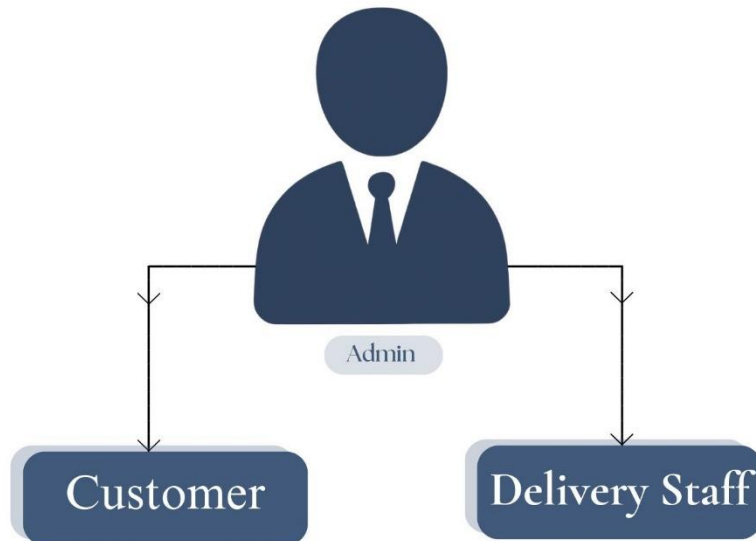
- **Strict Camera Privacy (Edge Computing):** The frontend must enforce HTTPS to request webcam permissions securely. The MediaPipe pose processing must execute entirely *locally* on the client's device. No video feeds, images, or biometric landmark data are ever to be transmitted to or stored on the Node.js server.
- **Data Protection:** The backend must utilize robust data protection, implementing JWT (JSON Web Tokens) for secure API sessions, strict CORS configurations to prevent cross-site request forgery, and bcrypt password encryption within the MongoDB database.

Usability & Compatibility:

- **Responsive UI:** The frontend must be styled entirely with Tailwind CSS, ensuring a mobile-first, fluid layout that adapts flawlessly to smartphones, tablets, and desktop monitors.
- **Cross-Browser WebGL Support:** The platform's 3D <Canvas>, <Environment>, and <ContactShadows> must be tested and fully functional across all modern WebGL-enabled browsers (Chrome, Safari, Firefox, Edge).

Proposed System

4.1 Scope



1. Admin / Owner Module

Serving as the platform's central command center, the Admin holds the highest level of authorization with absolute control over backend operations, 3D asset integration, inventory, and users:

- **Dashboard & Business Analytics:** Real-time visibility into vital business metrics, including total revenue, sales volume, customer acquisition costs, and crucially, monitoring the reduction in apparel return rates driven by the Virtual Try-On feature.
- **Catalog & 3D Asset Management (CRUD):** Full authority to add, edit, or delete clothing product listings. This includes uploading high-resolution 2D thumbnail images, defining specific apparel attributes (fabric, fit), and **managing the external GitHub raw URLs for the .glb 3D models** associated with each garment. Admins can also adjust base scaling parameters for the 3D models.

- **Complex Inventory & Order Management:** Complete oversight of the order lifecycle (payment verification, refunds, shipping tracking). The system allows the Admin to manage complex SKU variants, tracking stock levels across multiple sizes (S, M, L, XL) and colors to prevent overselling.
- **User & Access Control:** Manage all registered customer accounts and securely create restricted "Staff" roles. This allows the Owner to delegate daily fulfillment or customer service tasks without exposing sensitive financial data or master database controls.
- **Marketing & Promotions:** Configure platform-wide visual banners, generate promotional discount codes, and curate featured seasonal apparel collections (e.g., "Summer Lookbook" or "Limited Edition Drops").

2. Customer Module

Focused on delivering a frictionless, highly immersive, and premium shopping journey, this module restricts users to frontend interactions, personal data management, and the augmented reality fitting room:

- **Interactive Discovery & Virtual Try-On:** The core user experience. Customers can seamlessly browse the clothing catalog using advanced, dynamic filters (category, size, color, price) without page reloads. Most importantly, users can activate their device camera to utilize the **real-time 3D Virtual Try-On**, overlaying the .glb apparel models onto their bodies using the integrated MediaPipe AI tracking.
- **Profile & Sizing Management:** Securely register accounts using encrypted credentials, store multiple shipping/billing addresses, and save personalized sizing preferences to streamline future purchases.

- **Cart & Secure Checkout:** Manage a persistent React-state shopping cart, dynamically update quantities or sizes, apply promotional codes, and complete purchases via a secure, integrated, multi-step payment gateway.
- **Wishlists & Fit Reviews:** Curate personal wishlists for future consideration. Post-purchase, customers can submit ratings and reviews, specifically commenting on the real-world fit versus the 3D virtual representation, building community trust.
- **Order Tracking & History:** Access a dedicated user portal to view complete purchase histories, download digital invoices, and track active shipments in real-time.

4.2 Project Modules

The Thread Theory platform is divided into specialized functional modules to ensure a seamless user experience, secure backend operations, and the efficient rendering of complex 3D assets.

1. **User Registration & Authentication** Provides secure sign-up and login capabilities. The backend utilizes bcrypt for password hashing and JWT (JSON Web Tokens) for maintaining secure session states. Beyond standard profile management (storing encrypted billing and shipping addresses), this module allows users to save personalized sizing preferences and body metrics to streamline future purchases.
2. **Product Search & Catalog** Facilitates the dynamic browsing of the apparel collection. Users can utilize advanced, zero-reload filtering built in React (sorting by category, size, color, fabric, and price). Product pages serve as the primary visual hub, displaying traditional high-resolution 2D image galleries alongside detailed fabric care instructions and standard sizing charts.
3. **3D Virtual Try-On Engine (Core Innovation)** The standout module of the platform. Accessible directly from the product catalog, this feature requests secure webcam access. It utilizes @mediapipe/tasks-vision to process user skeletal landmarks locally in the browser. Simultaneously, @react-three/fiber fetches the associated .glb 3D clothing model from a GitHub raw link and maps it dynamically to the user's movements, providing a real-time, augmented reality fitting room experience.

4. **Shopping Cart & Wishlist** Customers can curate a persistent wishlist for future consideration and manage a dynamic, state-driven shopping cart. Built with React hooks, the cart updates instantly without page reloads. The checkout process provides transparent, real-time summaries detailing base apparel prices, applied promotional discounts, and calculated shipping costs.
5. **Payment & Billing** Integrates secure, third-party payment gateways (e.g., credit/debit cards, digital wallets, UPI) to handle financial transactions off-server, ensuring PCI compliance. Upon successful payment, the Node.js backend generates automated digital invoices detailing specific SKUs, sizes, and colors, while maintaining a complete, accessible transaction history for the user.
6. **Order & Fulfillment Management** Empowers customers to track their apparel shipments in real-time. On the backend, restricted Staff members utilize this module to manage active order queues, update fulfillment statuses (e.g., "Processing," "Quality Check," "Shipped"), generate digital packing slips, and input secure courier tracking numbers.
7. **Product & Asset Management (Admin)** Grants the platform Owner absolute CRUD (Create, Read, Update, Delete) authority over the master catalog. Crucially, this module is designed to handle complex SKU variants (managing stock levels across multiple sizes and colors) and includes dedicated input fields for Admins to link the external **GitHub raw URLs** and base scaling metrics for the 3D .glb assets.

8. **Dashboard & Analytics (Admin & Staff)** A comprehensive business intelligence interface. Admins can manage role-based access controls (RBAC) and view overarching metrics such as sales volume, revenue growth, and traffic. Most importantly, it tracks the platform's return rates, providing tangible data on how the Virtual Try-On feature is minimizing sizing-related returns. Staff dashboards focus on daily operational alerts, such as low-stock warnings for specific clothing variants.
9. **Review, Rating & Community Trust** Authenticated customers can submit verified star ratings and written reviews post-purchase. To maximize the value of this module for an apparel brand, users are encouraged to comment specifically on the real-world fit, fabric feel, and the accuracy of the 3D Virtual Try-On representation. Staff can monitor and moderate this feedback to maintain platform integrity.
10. **Customer Support & Returns** Features a detailed self-service FAQ covering shipping policies, fabric care, and advanced sizing guides. It includes a dedicated ticketing system that empowers Staff to efficiently track customer inquiries, manage size-exchange requests, and process authorized returns seamlessly.

4.3 Module-Wise Objectives, Functionalities, and Constraints

This section outlines the core modules of the Thread Theory platform, detailing their primary goals, key features, and the strict system limitations implemented to ensure smooth operation, secure data handling, and optimal rendering performance for the WebGL components.

1. User Authentication & Profile Management

- **Objective:** To provide secure, seamless access and a highly personalized experience tailored to individual shopper preferences and body metrics.
- **Functionalities:** Secure user registration, robust login/logout via JSON Web Tokens (JWT), profile detail updates, real-time order history tracking, wishlist management, and the ability to save personalized sizing and fit preferences.
- **Constraints:** All user passwords must be strictly encrypted using bcrypt prior to MongoDB storage. To ensure data security, JWT user sessions automatically expire after 24 hours of inactivity. Password reset requests require verified email link authentication.

2. Product Catalog & 3D Virtual Try-On (Core Module)

- **Objective:** To enable the intuitive discovery of fashion apparel and provide an immersive, augmented reality fitting room to reduce purchase hesitation and return rates.
- **Functionalities:** Dynamic product grid rendering via React, advanced zero-reload filtering (by category, size, color, and price), and detailed product view pages. Crucially, this module initializes the `@react-three/fiber` `<Canvas>` and activates `@mediapipe/tasks-vision` for real-time camera pose estimation and `.glb` model overlay.
- **Constraints:**
 - **2D Assets:** High-resolution 2D image uploads are restricted to a maximum of 5MB per file to maintain initial page load speeds.
 - **3D Assets:** To prevent server bandwidth overload, 3D `.glb` files **must not** be stored in the primary database. They must be hosted externally, and the platform will strictly fetch them via GitHub raw URLs.

- **Hardware Limits:** The Virtual Try-On feature requires the client device to have a functioning webcam, hardware acceleration enabled, and a WebGL-compatible browser.

3. Cart & Order Processing

- **Objective:** To facilitate a frictionless, error-free checkout process specifically designed to handle complex apparel variants (sizes and colors).
- **Functionalities:** Real-time, state-driven cart updates, dynamic SKU variant management (ensuring the user selects both size and color before adding to cart), secure third-party payment gateway integration, order confirmation, and automated invoice generation.
- **Constraints:** Only one promotional discount coupon can be applied per order (no stacking). To manage high-demand inventory, adding an item to the cart does *not* deduct it from the master database stock; inventory is only officially reserved upon successful payment authorization.

4. Admin Dashboard, Inventory & Asset Management

- **Objective:** To provide a centralized, powerful, and secure interface for store administrators to manage the dynamic apparel catalog and monitor business health.
- **Functionalities:** Complete CRUD operations for the clothing catalog. This includes managing complex stock arrays (e.g., tracking how many "Medium" vs. "Large" shirts are available), processing refunds, and tracking shipment statuses. **Crucially, the Admin must have input fields to paste external GitHub .glb URLs and adjust 3D base-scaling modifiers for each product.**
- **Constraints:** Access is strictly limited to accounts with verified "Admin" or "Staff" role privileges via secure routing middleware. To prevent database query overload, heavy business intelligence analytics (like total revenue or traffic reports) are cached and updated in 5-minute intervals rather than triggering continuous database aggregation pipelines.

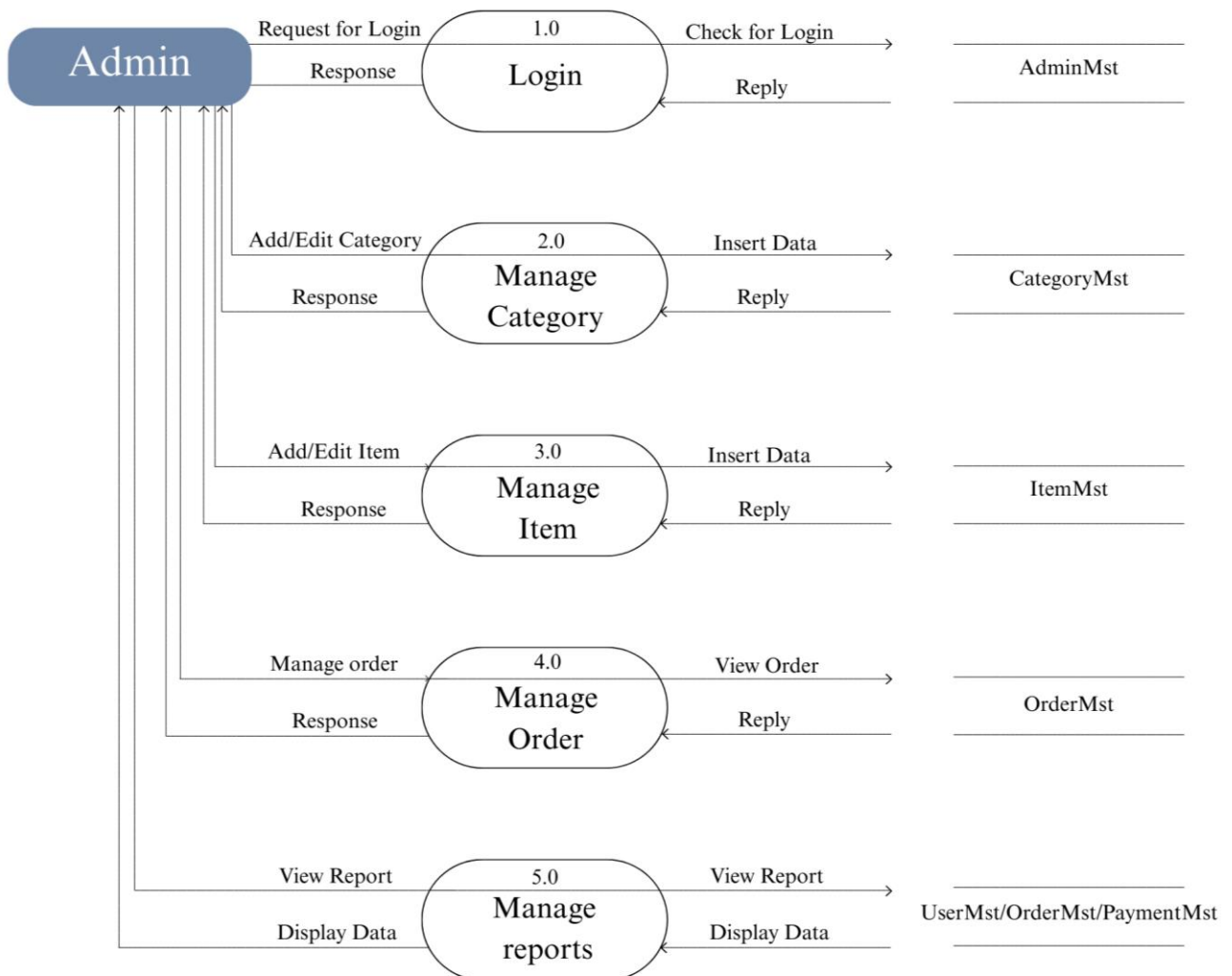
Detail Planning

5.1 Data Flow Diagram

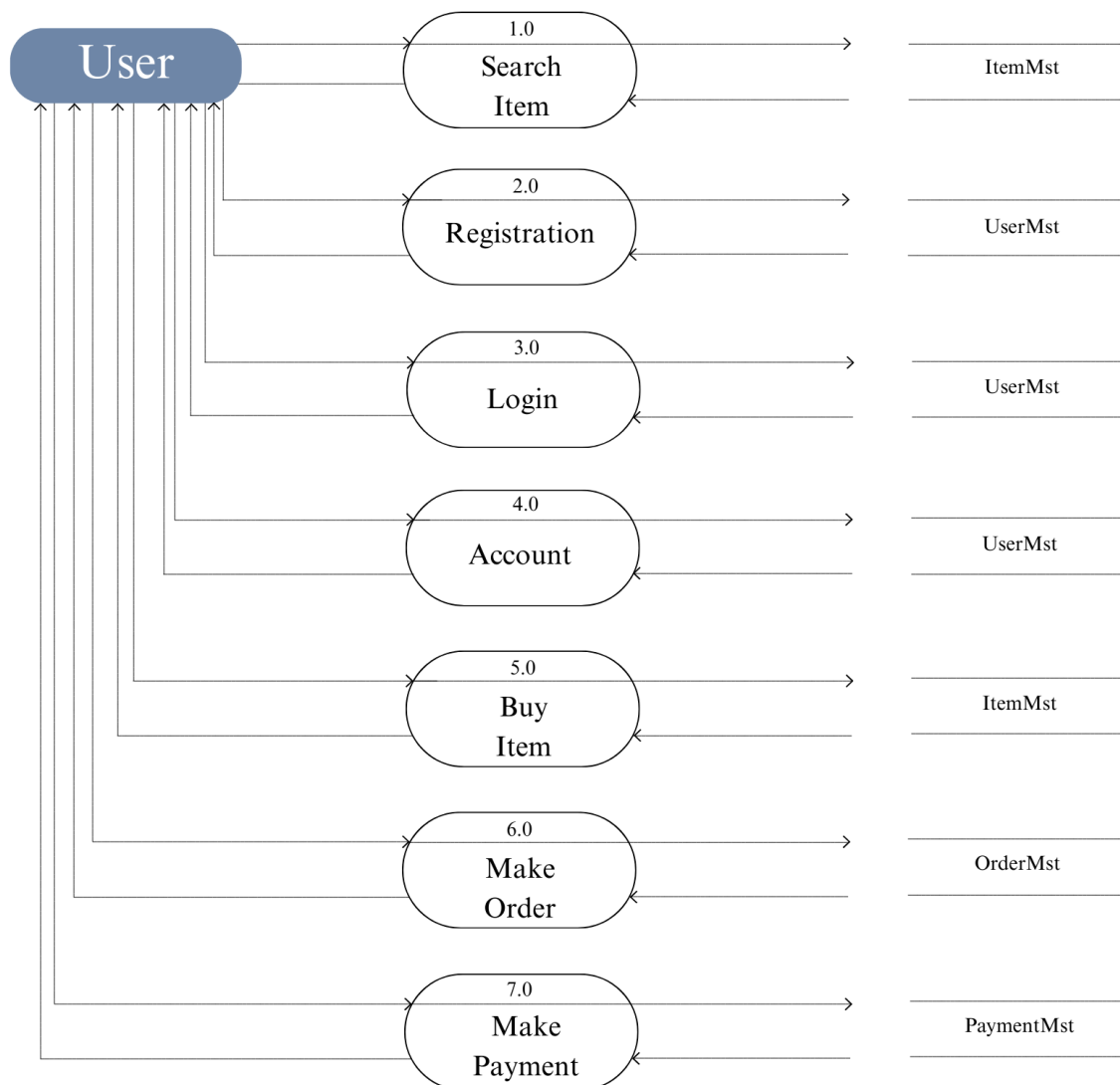
Context-level



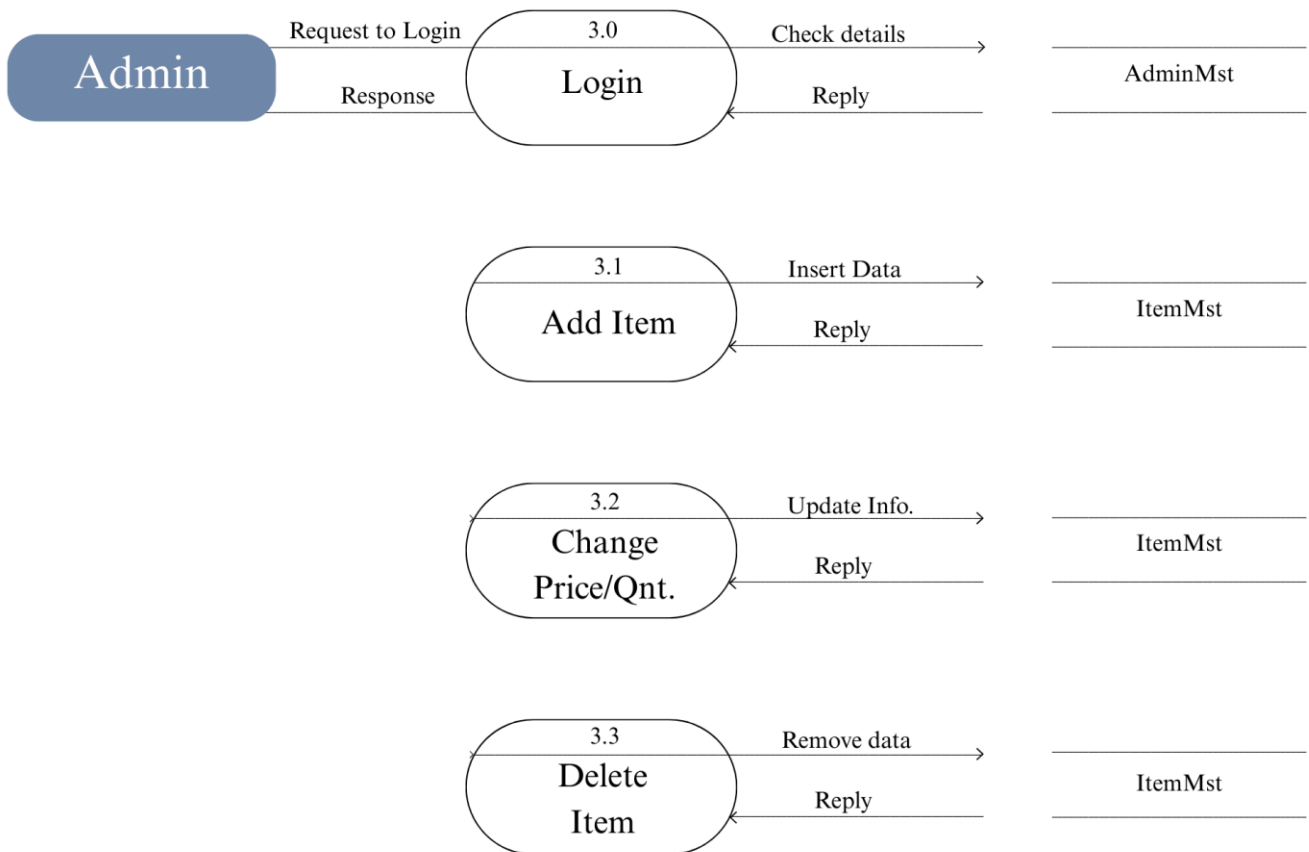
Admin side DFD - 1st level



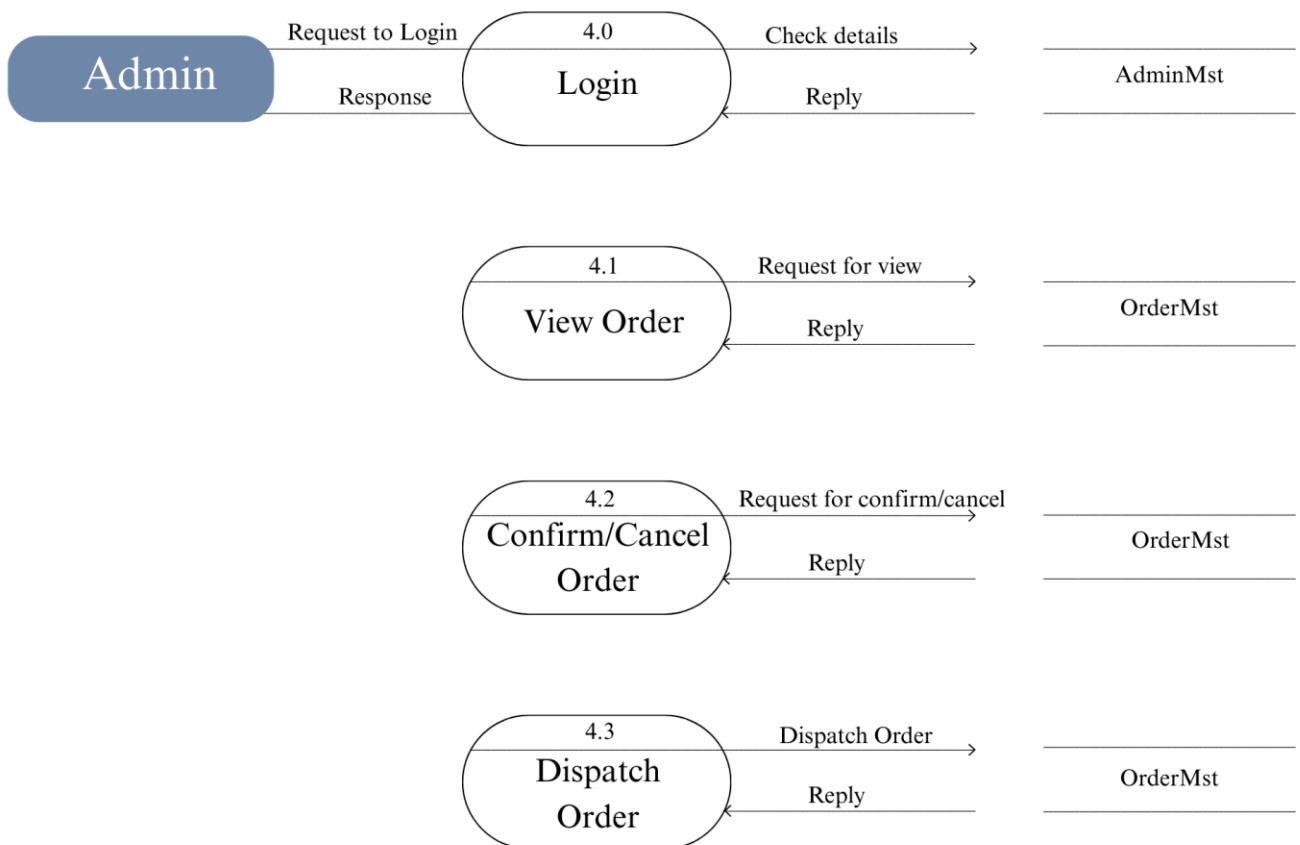
User side DFD - 1st level



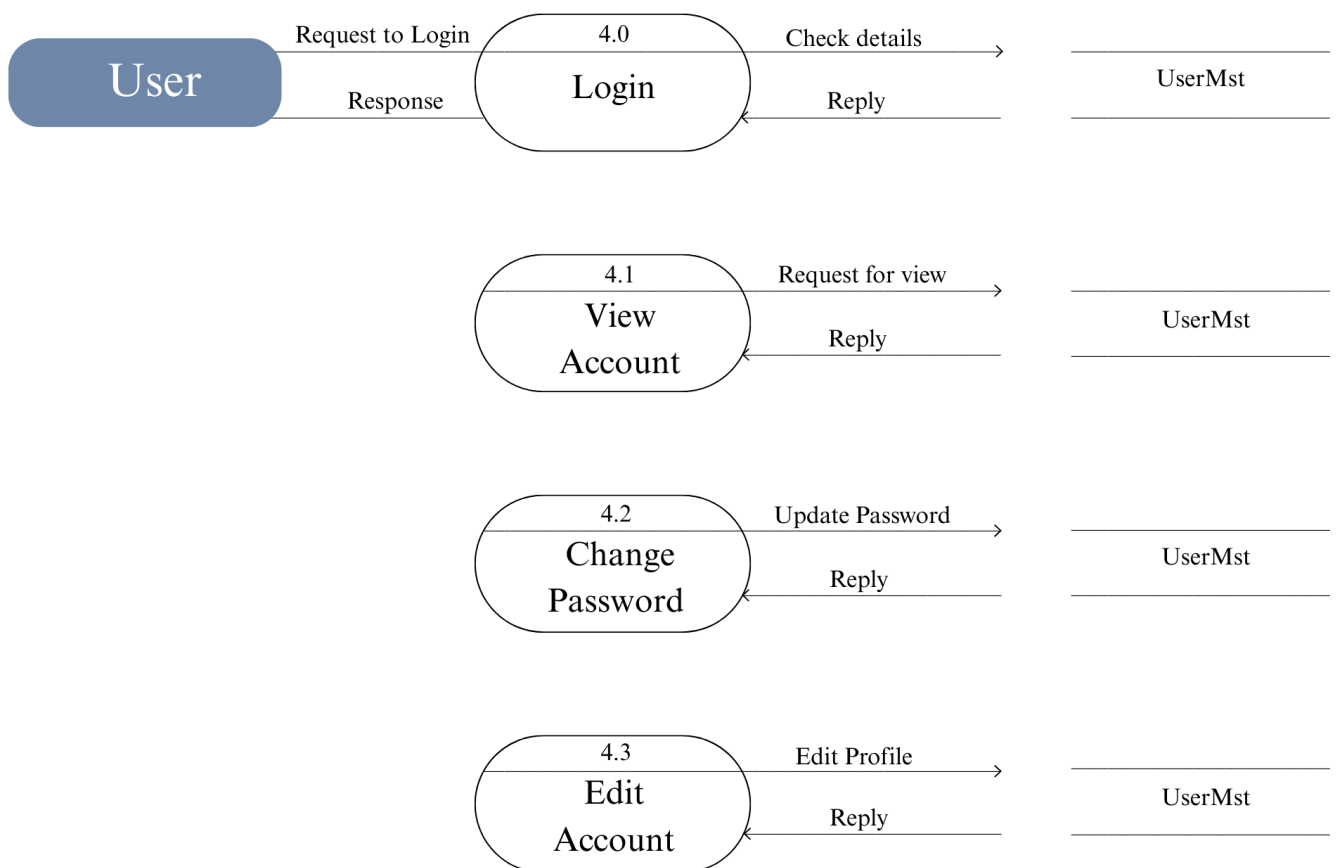
Admin side DFD - 2nd level [3.0]



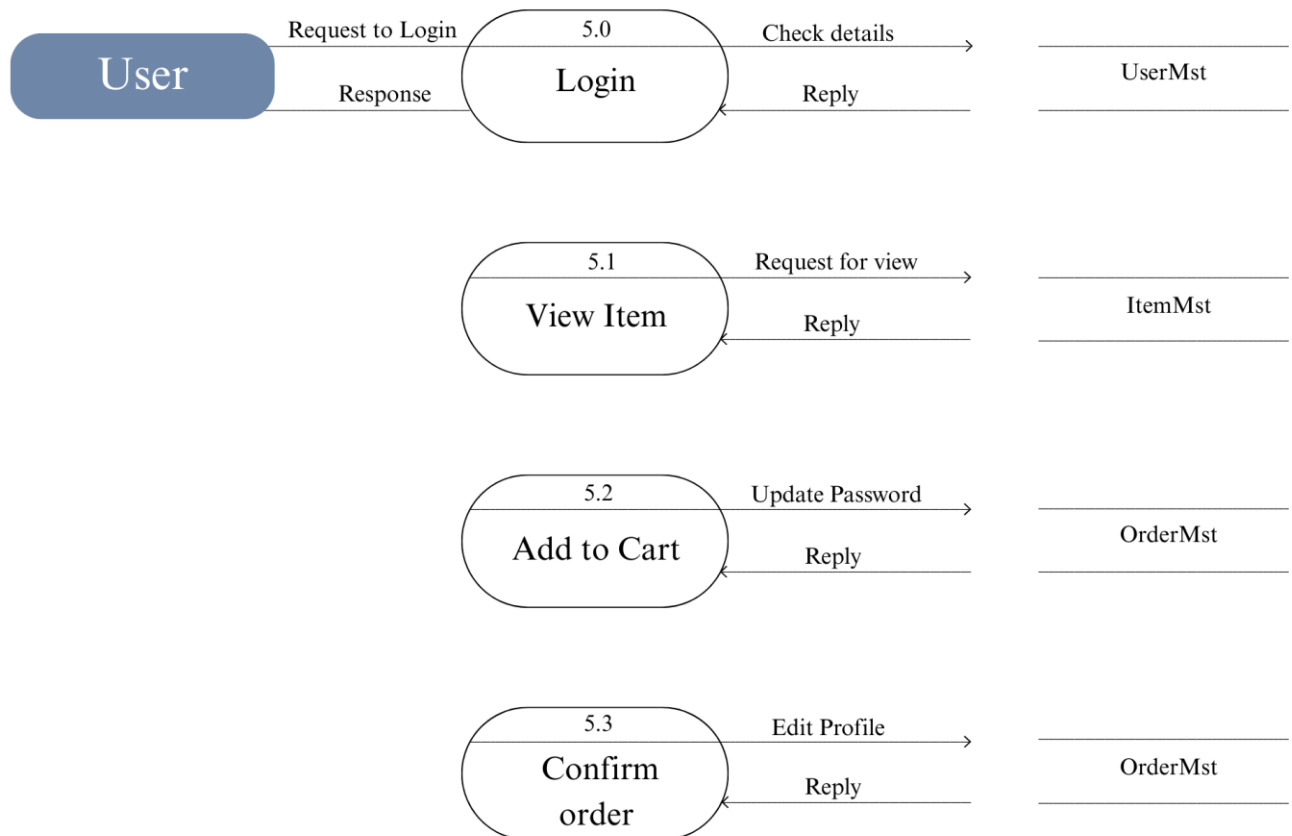
Admin side DFD - 2nd level [4.0]



User side DFD - 2nd level [4.0]

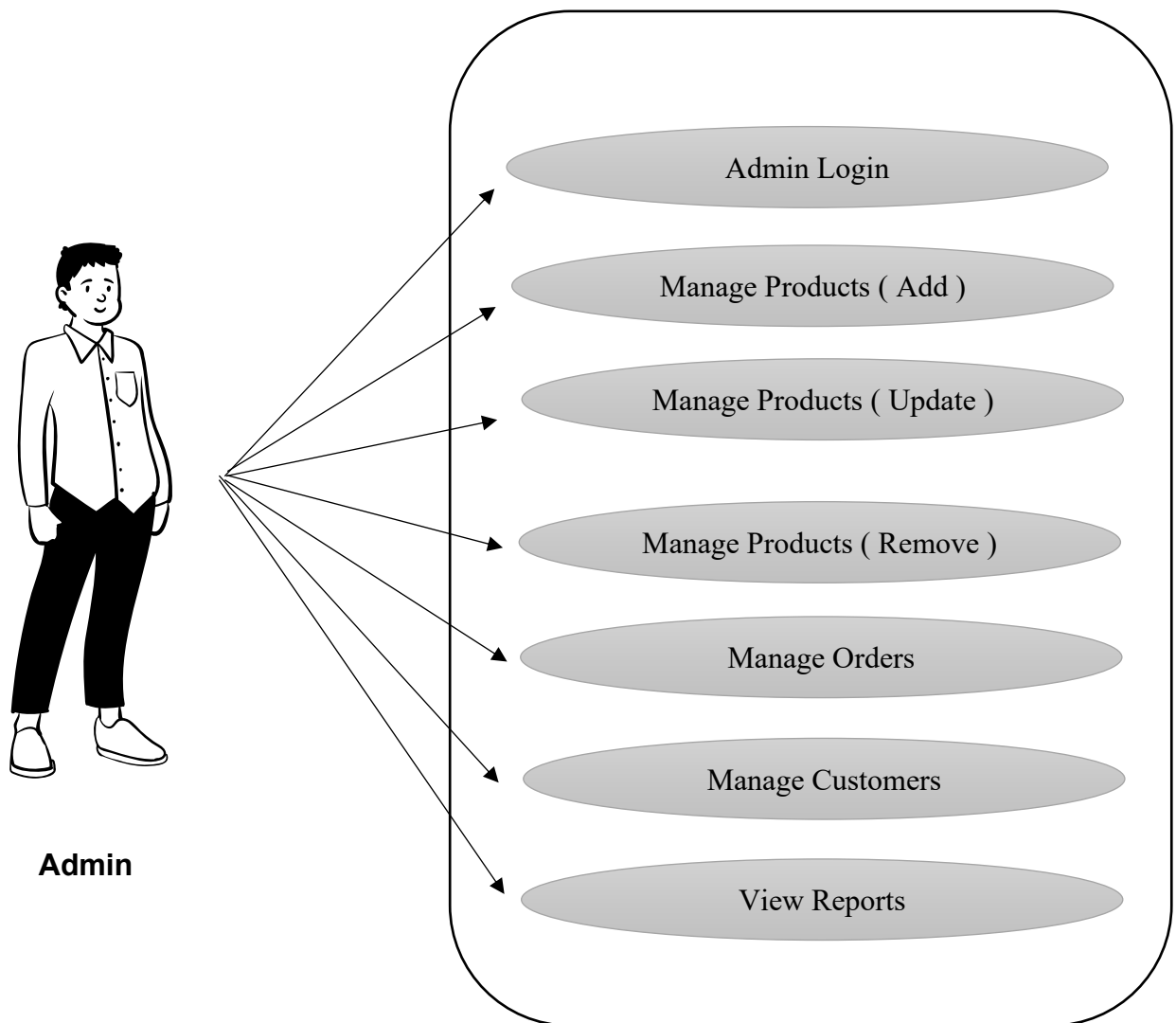


User side DFD - 2nd level [5.0]

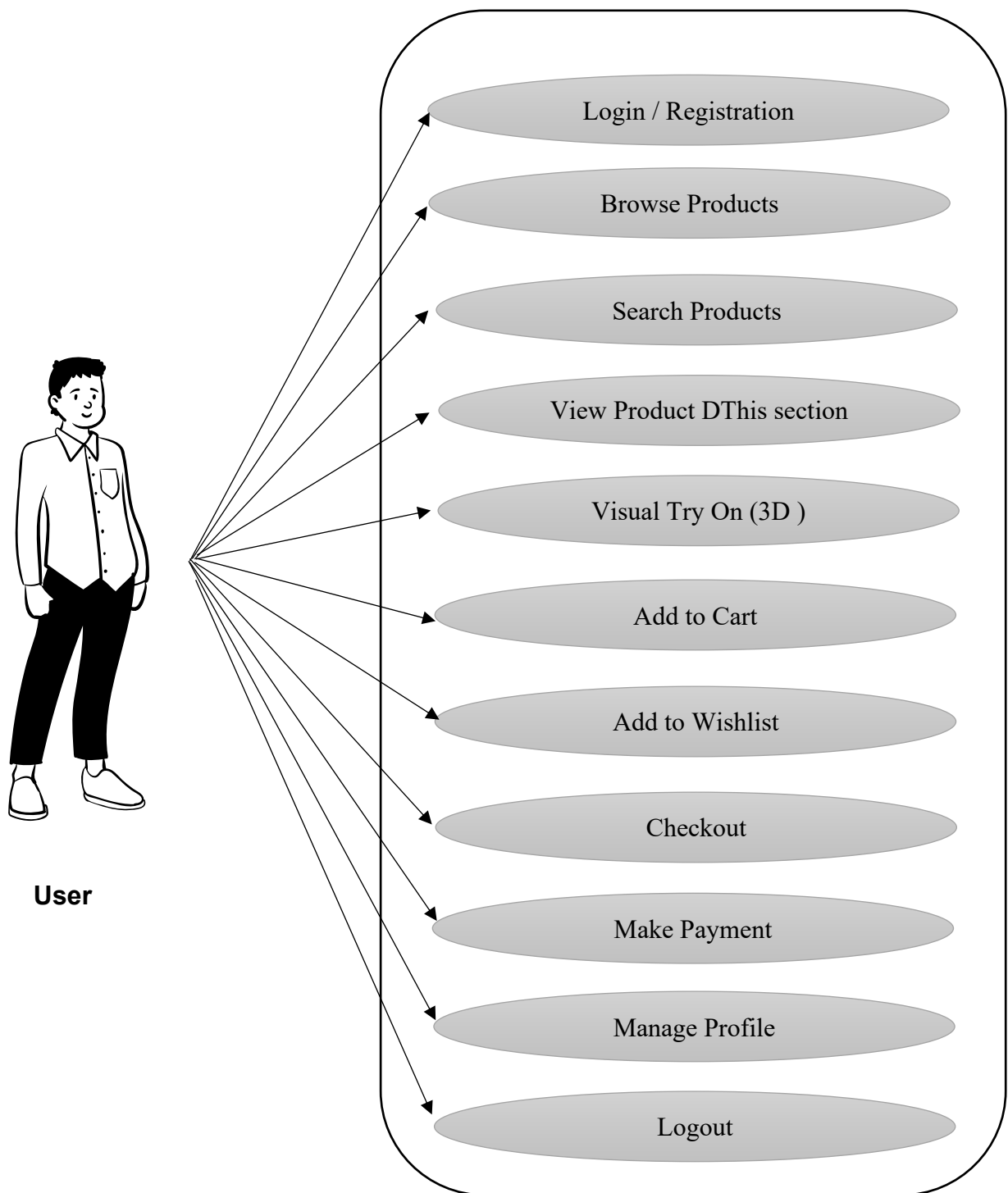


5.2 UseCase Diagram

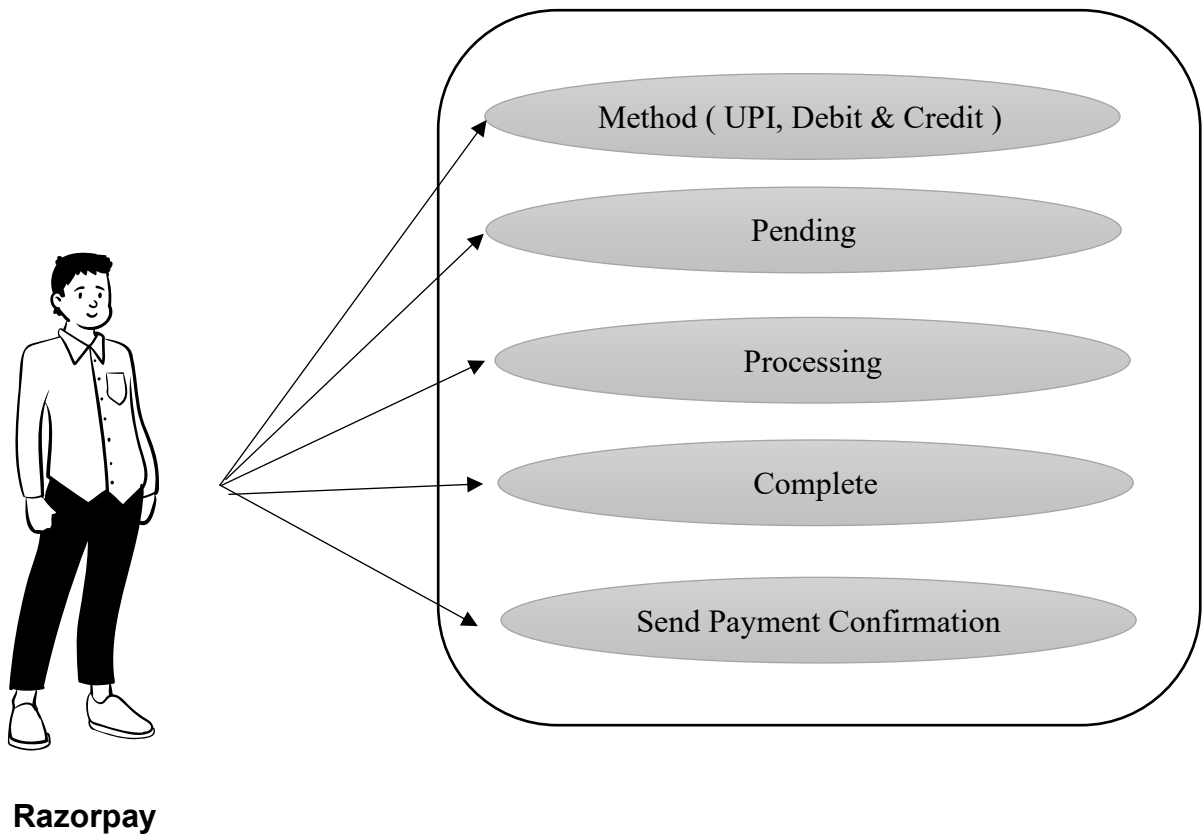
❖ Admin Use Case Diagram



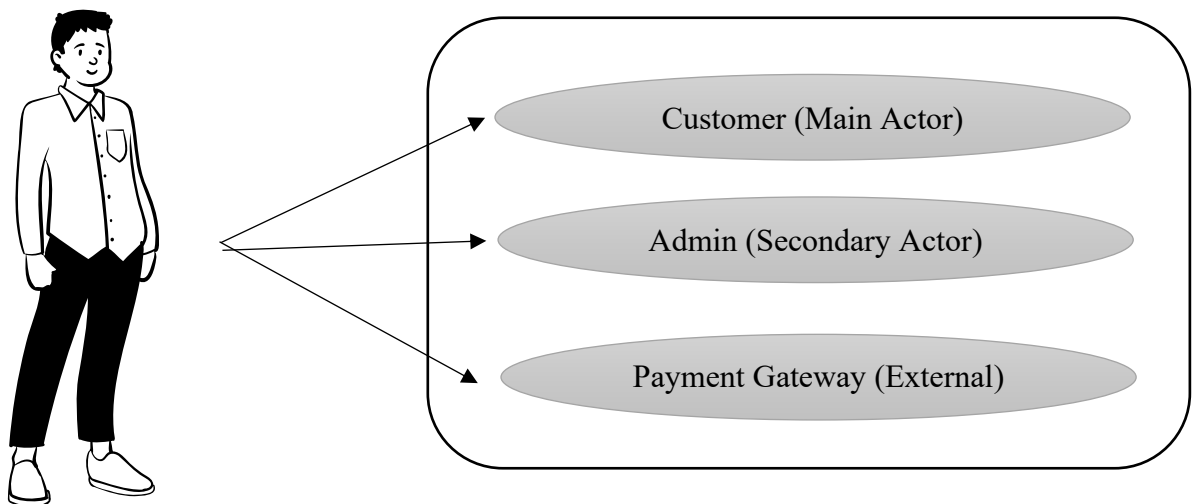
❖ Customer Use Case Diagram



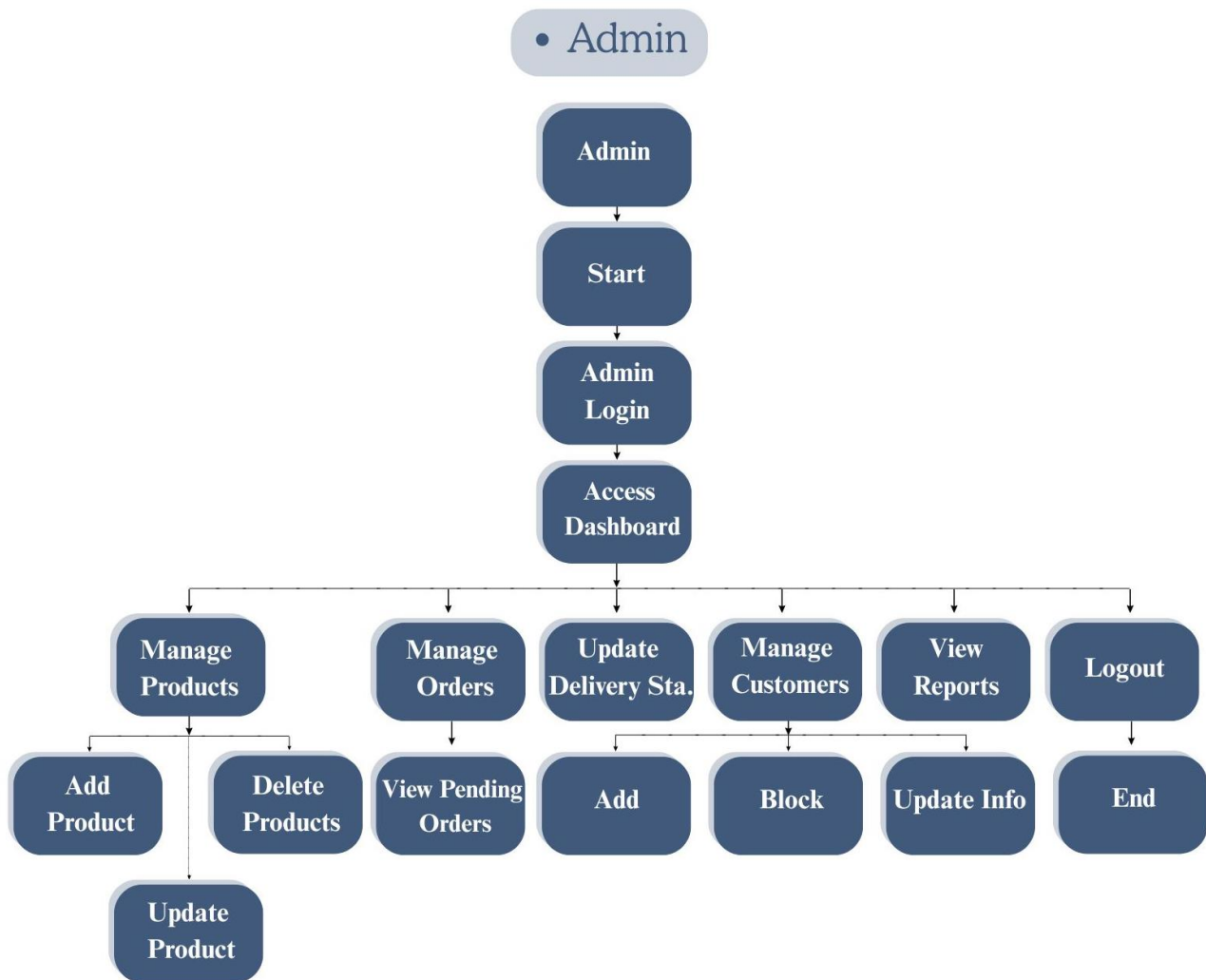
❖ Payment Diagram

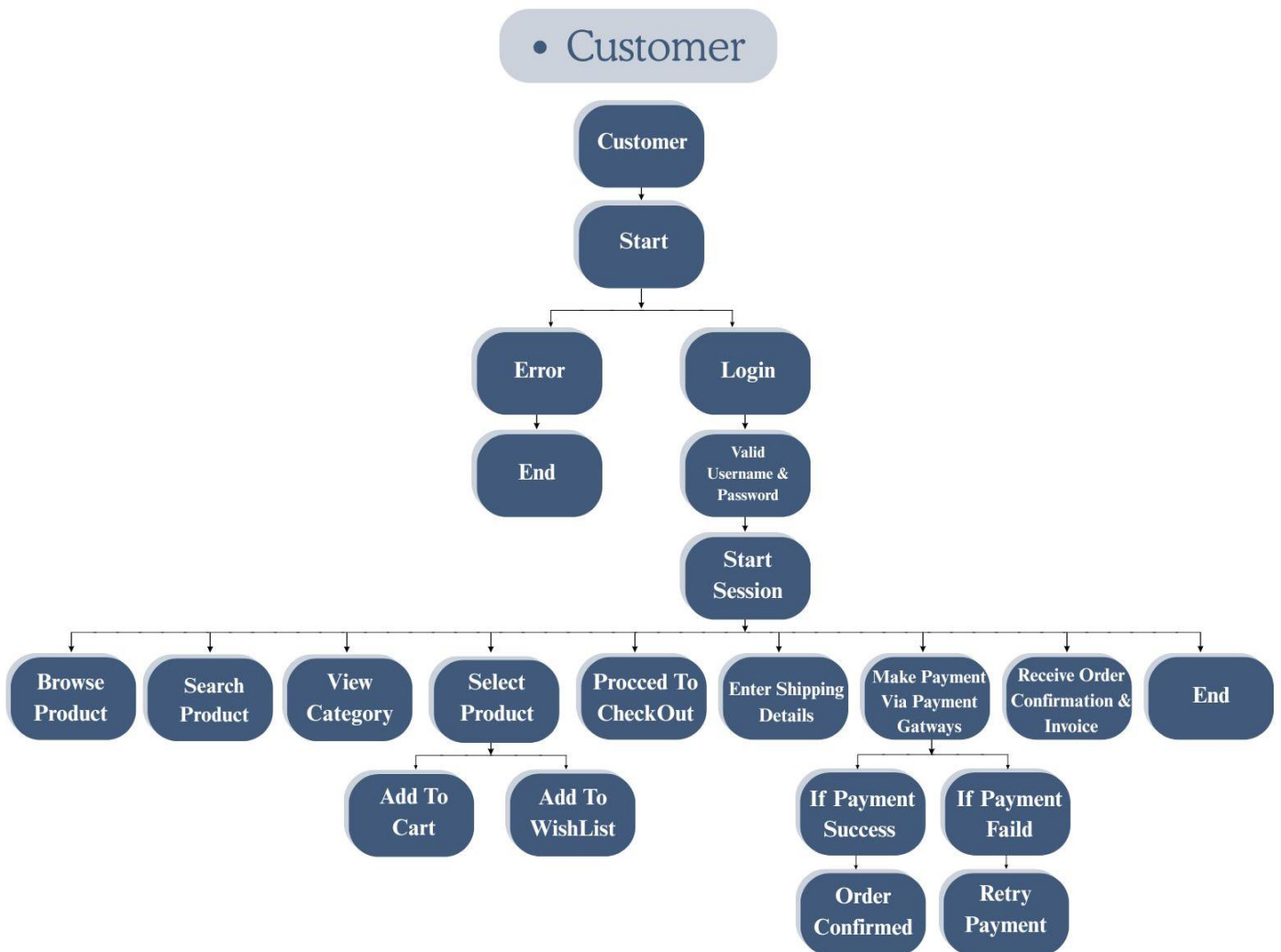


❖ Actor Diagram

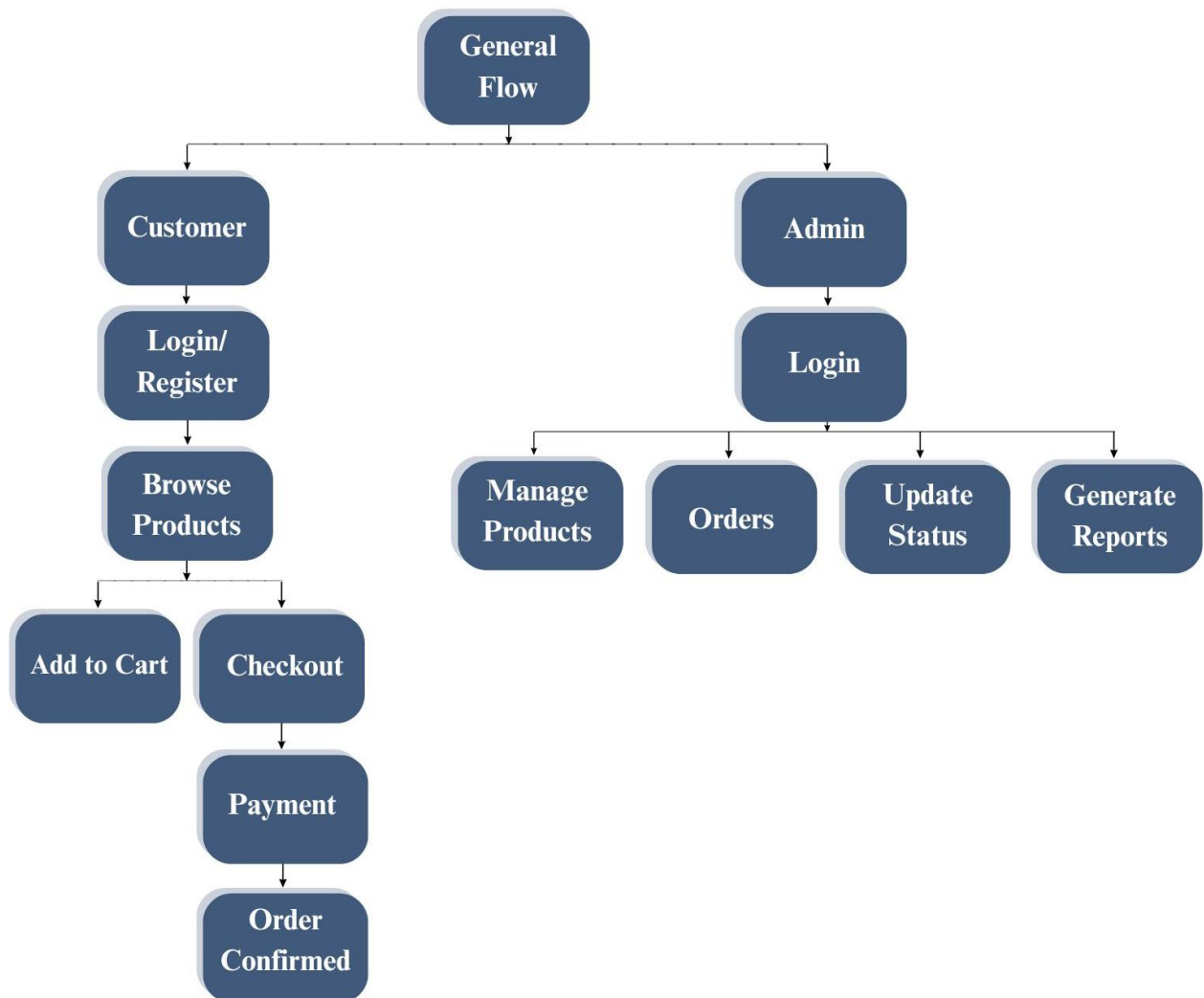


5.3 Activity Flow Diagram

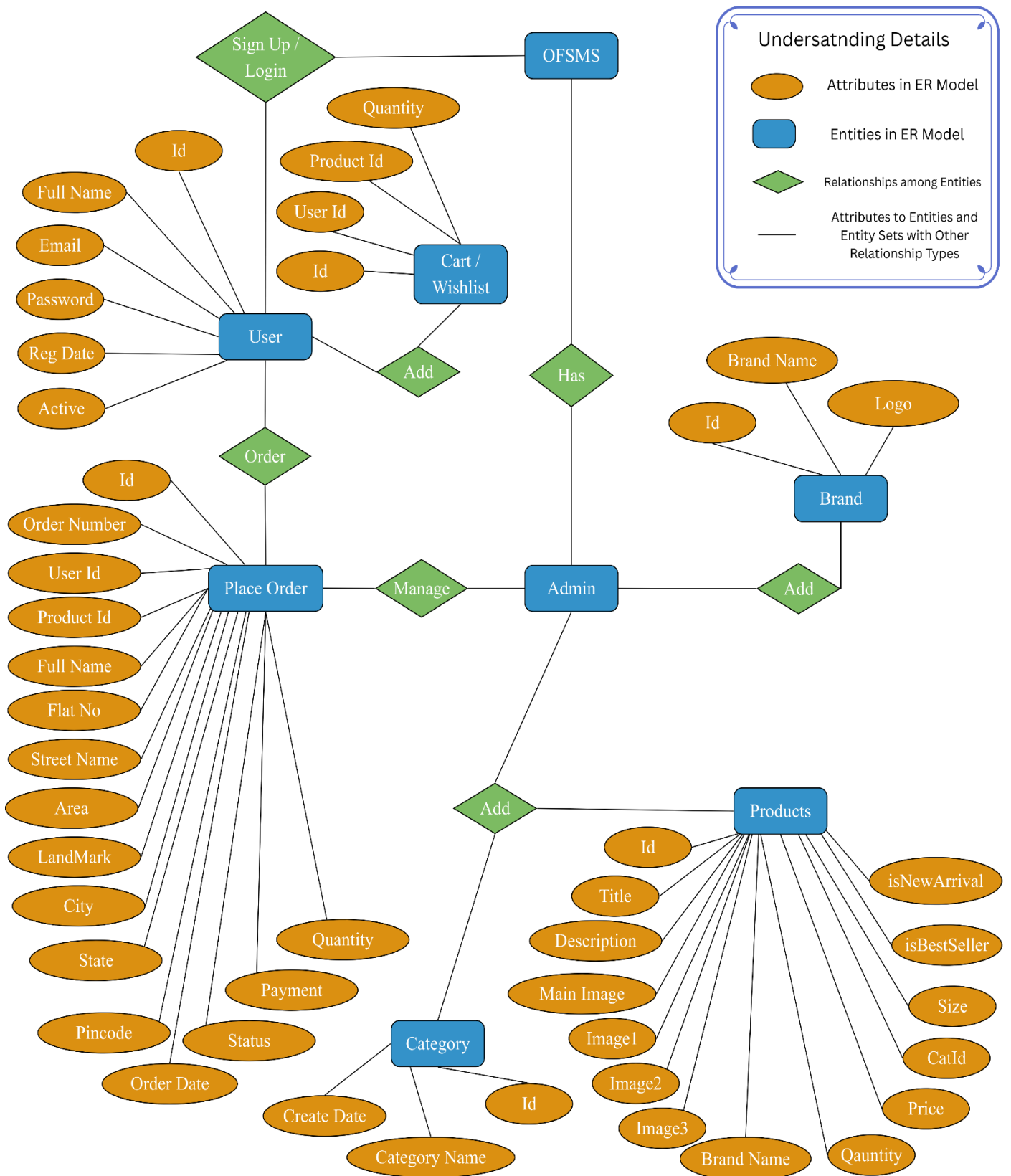




- General Flow Representation



5.4 Entity-Relationship Diagram / Class Diagram



System Design

6.1 Database Design

6.1.1. User Table

Stores all client and administrator account information, including secure authentication and delivery details.

Field Name	Description	Data Type
_id	Unique identifier for each user	ObjectId
name	Full name of the user	String
email	Email address of the user (Unique)	String
password	Encrypted/Hashed password for authentication	String
phone	Contact number for delivery	String
address	Object containing street, city, state, zip, country	Object
role	Defines user privileges (user or admin)	String
isBlocked	Indicates if the account is suspended by an admin	Boolean
resetPasswordToken	Secure token for password recovery	String
resetPasswordExpire	Expiration timestamp for the recovery token	Date
createdAt	Account creation timestamp	Date
updatedAt	Last profile update timestamp	Date

6.1.2. Cart Table

Manages the temporary shopping bag items for users before they proceed to checkout.

Field Name	Description	Data Type
_id	Unique cart identifier	ObjectId
userId	Reference to the user	ObjectId
items	Array of products (includes productId, size, qty)	Array
createdAt	Cart initialization date	Date
updatedAt	Last time an item was added/removed/updated	Date

6.1.3. Wishlist Table

Stores curated/saved pieces that the user intends to view or purchase later.

Field Name	Description	Data Type
_id	Unique wishlist identifier	ObjectId
userId	Reference to the user	ObjectId
products	Array of productId references saved by the user	Array
createdAt	Wishlist creation date	Date

6.1.4. Product Table

Stores all clothing and accessory inventory details, media assets, and stock levels

Field Name	Description	Data Type
_id	Unique identifier of the product	ObjectId
title	Name/Title of the clothing piece	String
brand	Brand or designer name	String
description	Detailed description of the fabric, fit, and care	String
price	Selling price of the item	Number
category	Target category (Men, Women, Kids, Accessories)	String
season	Collection season (Summer, Winter, All)	String
sizesAvailable	Array of available sizes (e.g., ['S', 'M', 'L'])	Array
stock	Current inventory quantity available for purchase	Number
mainimage1	Primary thumbnail image URL	String
image2	Secondary image URL (Hover state)	String
image3, image4	Additional product gallery image URLs	String
model3Durl	Link to AR/3D model file (GLTF/GLB)	String
createdAt	Product addition timestamp	Date

6.1.5. Order Table

Tracks all customer purchases, shipping addresses, total amounts, and fulfillment phases.

Field Name	Description	Data Type
_id	Unique identifier of the order (Order Ref)	ObjectId
userId	Reference to the user who placed the order	ObjectId
products	List of items bought (includes productId, size, qty)	Array
shippingAddress	Delivery address snapshot at the time of order	Object
paymentMethod	Payment mode used (e.g., Card, UPI, COD)	String
paymentStatus	Current payment state (PAID, UNPAID, REFUNDED)	String
totalAmount	Final payable/paid amount	Number
orderStatus	Fulfillment state (Pending, Processing, Shipped...)	String
createdAt	Date and time the order was placed	Date
updatedAt	Last status update timestamp	Date

6.1.6. Returns & Disputes Table

Manages user return requests and administrative approval/rejection workflows.

Field Name	Description	Data Type
_id	Unique return request identifier	ObjectId
orderId	Reference to the original order	ObjectId
userId	Reference to the client making the return	ObjectId
productId	Reference to the specific item being returned	ObjectId
quantity	Number of units being returned	Number
reason	Customer's selected reason for return (e.g., "Size Issue")	String
comments	Additional context provided by the customer	String
status	Request state (Pending, Approved, Refunded, Rejected)	String
adminNote	Explanatory message sent from admin to customer	String
createdAt	Date the return was requested	Date

6.1.7. Coupons Table

Manages promotional codes, discount rules, and usage tracking for marketing campaigns.

Field Name	Description	Data Type
_id	Unique coupon identifier	ObjectId
code	The actual promo code (e.g., SUMMER20)	String
discountType	Type of discount (percentage or fixed)	String
discountValue	The value to deduct (e.g., 20 for 20% or ₹500)	Number
minPurchase	Minimum cart total required to apply the coupon	Number
expiryDate	Exact date and time the coupon becomes invalid	Date
usageLimit	Maximum number of times this coupon can be used total	Number
usedCount	Running total of how many times it has been applied	Number
isActive	Manual toggle to turn the coupon on/off	Boolean
createdAt	Date the coupon was generated	Date
updatedAt	Last modification date	Date

6.1.8. Product History Table (Audit Trail)

Maintains a strict ledger of what changes were made to the catalog and which administrator made them.

Field Name	Description	Data Type
_id	Unique history record identifier	ObjectId
productId	Reference to the specific product altered	ObjectId
changedBy	Reference to the Admin userId who made the edit	ObjectId
action	Type of change (Created, Updated, PriceDrop, Restock)	String
changesMade	Object detailing previous values vs. new values	Object
createdAt	Exact timestamp of the administrative action	Date

6.1.9. Payment Ledger Table

Stores raw transaction data from payment gateways (like Stripe, Razorpay, or PayPal). This is crucial for financial reconciliation and handling refunds.

Field Name	Description	Data Type
_id	Unique internal payment identifier	ObjectId
orderId	Reference to the Thread Theory Order	ObjectId
userId	Reference to the client making the payment	ObjectId
transactionId	External ID provided by the payment gateway	String
paymentGateway	Service used (e.g., Razorpay, Stripe)	String
amountPaid	Exact amount charged to the customer	Number
currency	Currency code (e.g., INR, USD)	String
paymentStatus	Gateway status (Success, Failed, Refunded)	String
receiptUrl	Link to the generated invoice/receipt	String
createdAt	Timestamp of the transaction attempt	Date
updatedAt	Timestamp of status changes (like refunds)	Date

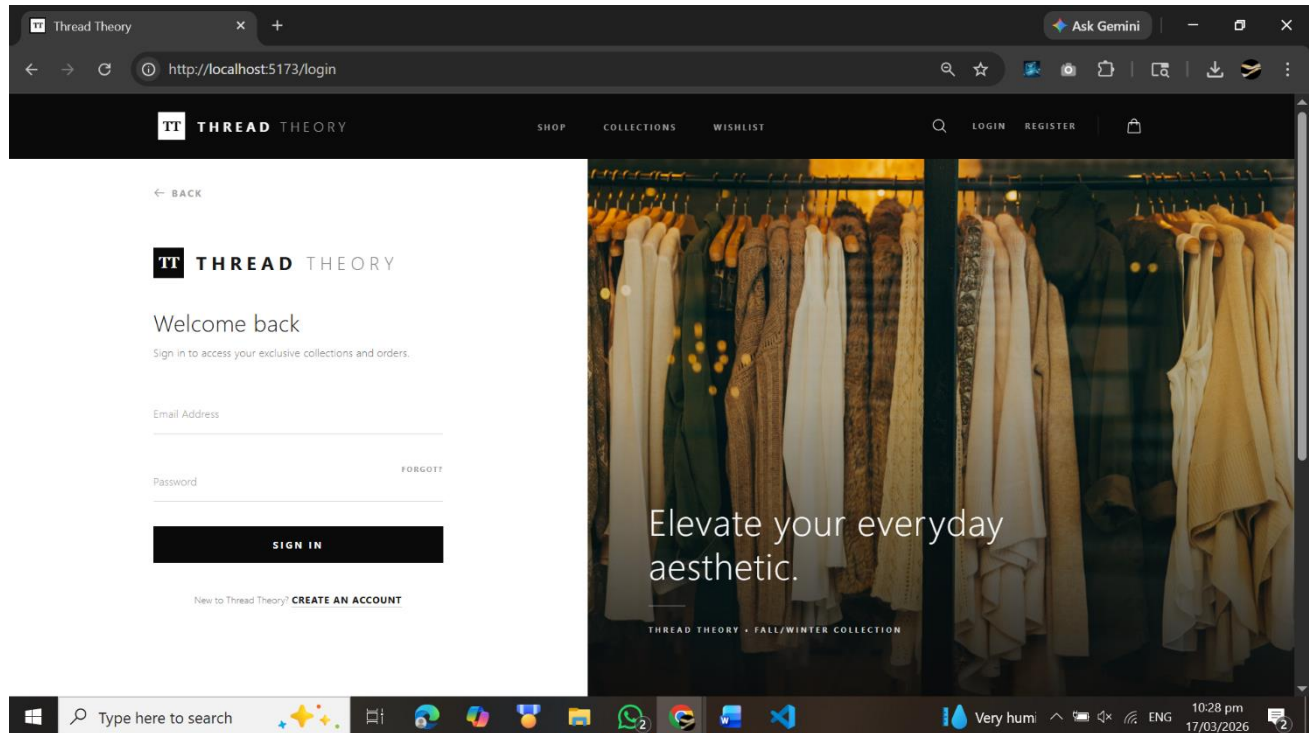
6.1.10. Product Reviews Table

Captures client feedback, ratings, and optional user-uploaded images for specific clothing pieces.

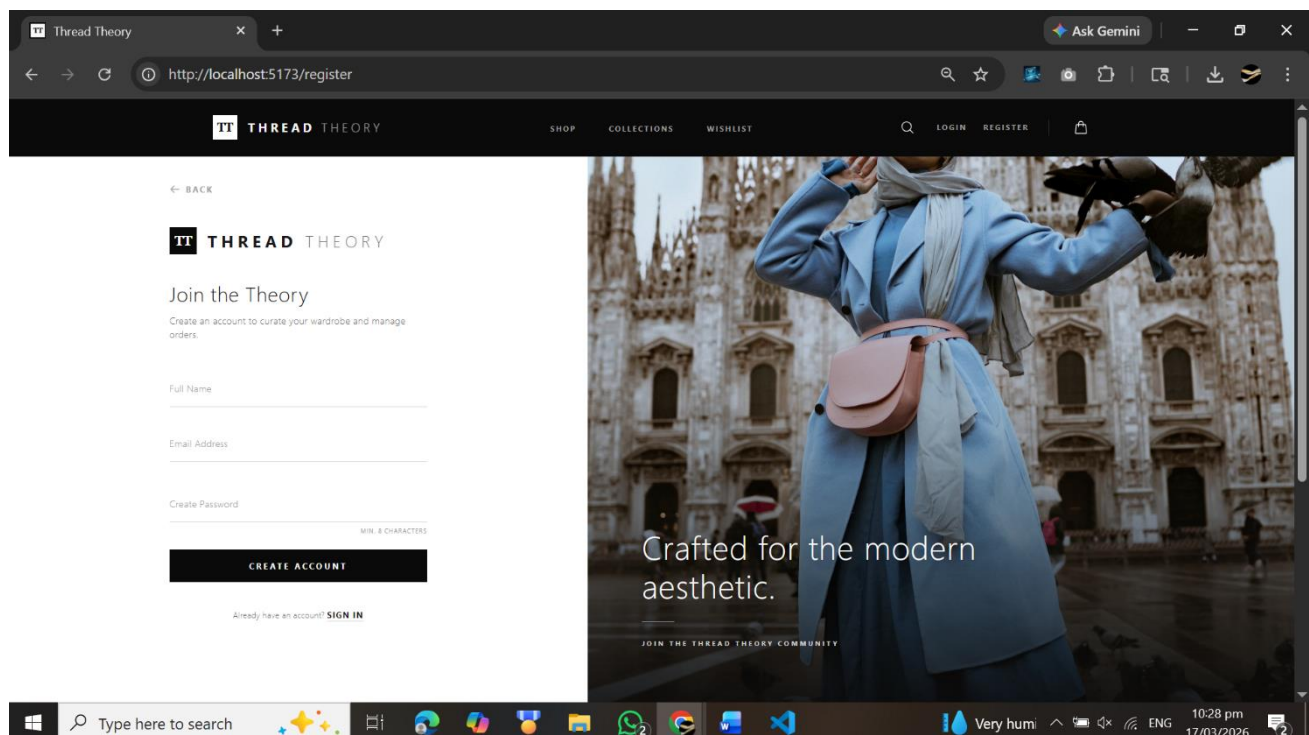
Field Name	Description	Data Type
_id	Unique review identifier	ObjectId
productId	Reference to the reviewed product	ObjectId
userId	Reference to the client who wrote the review	ObjectId
rating	Star rating (Number between 1 and 5)	Number
title	Brief summary/headline of the review	String
comment	Full text of the customer's feedback	String
images	Array of image URLs uploaded by the client	Array
isVerifiedPurchase	Boolean confirming the user actually bought the item	Boolean
isApproved	Admin toggle to approve/hide reviews for moderation	Boolean
createdAt	Date the review was submitted	Date
updatedAt	Date the review was last edited by the user	Date

6.2 User Interface Design

Login Page :



Registration Page :



Forgot Password :

TT

THREAD THEORY

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COLLECTIONS

WISHLIST

Q

LOGIN

REGISTER



← BACK TO LOGIN

RESET PASSWORD

ENTER YOUR EMAIL TO RECEIVE RECOVERY INSTRUCTIONS.

EMAIL ADDRESS

SEND RECOVERY LINK

Reset Password Link Send :

TT

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LOGIN

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← BACK TO LOGIN

RESET PASSWORD

ENTER YOUR EMAIL TO RECEIVE RECOVERY INSTRUCTIONS.



If an account exists for **smruti@gmail.com**, a recovery link has been generated.

DEVELOPMENT MODE BYPASS

[CLICK HERE TO RESET](#)

New Password Update :

TT

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LOGINREGISTER

← BACK TO LOGIN

NEW PASSWORD

PLEASE ENTER YOUR NEW SECURITY CREDENTIALS.

NEW PASSWORD

8+ CHARS • A-Z • a-z • 0-9 • SYMBOL

CONFIRM PASSWORD

SECURE ACCOUNT

Admin Pages

Product Page :


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7


2


3


2


0


ADD NEW PIECE

INVENTORY ARCHIVE

CATALOG MANAGEMENT (7 PIECES)

ITEM	DETAILS	INVENTORY	ACTIONS
	Men Winter Jacket by Gucci MEN • WINTER	₹3,499 4 IN STOCK	
	Women Kurti by ZARA WOMEN • SUMMER	₹599 4 IN STOCK	
	Women Jeans by H&M WOMEN • WINTER	₹1,799 15 IN STOCK	
	Men's Formal Down Shirt for Men by Lacoste MEN	₹599 18 IN STOCK	
	The Cotton Pro Tee by Nike MEN • SUMMER	₹1,499 46 IN STOCK	
	Jacquard-Knit Cotton Jumper by Givenchy KIDS	₹899 10 IN STOCK	
	Green Typographic Print Polo Neck T-Shirt by Armani Exchange KIDS • WINTER	₹1,299 OUT OF STOCK	


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ADD product :

CURATE NEW PIECE

TITLE

BRAND *

PRICE (₹)

STOCK QTY

CATEGORY

SEASON

Select Category

Select Season

SIZES (S, M, L...)

DESCRIPTION

Product details...

MEDIA ASSETS

MAIN IMAGE 1 *

IMAGE 2 (HOVER)

IMAGE 3

IMAGE 4

3D MODEL URL (GLTF/GLB)

PUBLISH PIECE

CURATE NEW PIECE

TITLE

BRAND *

PRICE (₹)

STOCK QTY

CATEGORY

SEASON

Women

All

SIZES (S, M, L...)

S, M, L, XL, XXL, 3XL

DESCRIPTION

for women season.
Size Recommendation: Please use the infographic size chart guidance to determine your perfect size for office wear tops for women ; a size large is recommended for a comfortable fit.

MEDIA ASSETS

MAIN IMAGE 1 *

IMAGE 2 (HOVER)

IMAGE 3

IMAGE 4

<https://m.media-amazon.com/images/I/>
<https://m.media-amazon.com/images/I/>

<https://m.media-amazon.com/images/I/>
<https://m.media-amazon.com/images/I/>

PUBLISH PIECE

Order Page :


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[USERS](#)
[ANALYTICS](#)
[STOCKOUTS](#)
[RETURNS](#)

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LOGISTICS

THREAD THEORY - FULFILLMENT CONTROL

TODAY

PENDING
1

PROCESSING
1

SHIPPED
0

DELIVERED
0

RETURNED
0

ALL SHIPMENTS

ORDER REF	CLIENT	MANIFEST	TOTAL	STATUS
#F46B59	Fenil Patel fenil@gmail.com		₹3,998.4	PROCESSING MAR 17, 2026, 10:35 PM
#5F3280	Payal Patel payal@gmail.com		₹1,099	PENDING MAR 17, 2026, 11:18 AM


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[SYSTEM CORE](#)

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[PRODUCTS](#)

[ORDERS](#)

[COUPONS](#)

[USERS](#)

[ANALYTICS](#)

[STOCKOUTS](#)

[RETURNS](#)

CLIENT DIRECTORY

THREAD THEORY - ACCESS & PRIVILEGES

TOTAL CLIENTS

10

ACTIVE

9

SUSPENDED

1

ADMINISTRATORS

2

ALL RECORDS

CLIENT INFO	CONTACT	ROLE	STATUS	ACTIONS
 ronak	ronak@gmail.com	ADMIN	● ACTIVE	
 nand	nand@gmail.com	ADMIN	● ACTIVE	
 jash	jash@gmail.com	USER	● SUSPENDED	
 Manthan	manthan@gmail.com	USER	● ACTIVE	
 Fenil Patel	fenil@gmail.com	USER	● ACTIVE	
 Smit Pokar	smit23@gmail.com	USER	● ACTIVE	
 Smruti Patel	smruti@gmail.com	USER	● ACTIVE	
 Payal Patel	payal@gmail.com	USER	● ACTIVE	
 Heet Patel	heet@gmail.com	USER	● ACTIVE	
 Meet	meet@gmail.com	USER	● ACTIVE	

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Promotions Page Coupon :

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Q U B

TT ADMIN

SYSTEM CORE

DASHBOARD

PRODUCTS

ORDERS

COUPONS

USERS

ANALYTICS

STOCKOUTS

RETURNS

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PROMOTIONS ARCHIVE

THREAD THEORY - INCENTIVE MANAGEMENT

GENERATE CODE

CODE NAME

DISCOUNT (%)

EXPIRATION DATE

dd/mm/yyyy

ACTIVATE CODE

ACTIVE & PAST REGISTRY

CODE	VALUE	STATUS	EXPIRES	ACTION
H0H26	40% OFF	EXPIRED	MAR 3, 2026	
FIRST20	20% OFF	ACTIVE	MAR 25, 2026	
SDJ50	50% OFF	EXPIRED	MAR 1, 2026	
WELCOME20	20% OFF	ACTIVE	DEC 31, 2026	

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[ORDERS](#)

[COUPONS](#)

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ALL DEPLETED

1

WOMEN

0

MEN

0

KIDS

1

ACCESSORIES

0

FULL ALERT LIST

ITEM	DETAILS	STATUS	ACTION
	Green Typographic Print Polo Neck T-Shirt KIDS • WINTER	OUT OF STOCK	RESTOCK

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[SYSTEM CORE](#)
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RETURN REQUESTS

THREAD THEORY - DISPUTE MANAGEMENT

PENDING
0

APPROVED
2

REFUNDED
1

REJECTED
0

ALL REQUESTS

ORDER	ITEM & CLIENT	REASON	STATUS
#401E3F MAR 9, 2026	 Men Winter Jacket QTY: 1 • MEET	CHANGED MY MIND	APPROVED
#38ED71 MAR 5, 2026	 Jacquard-Knit Cotton Jumper QTY: 1 • PAYAL PATEL	SIZING ISSUE - TOO LARGE	REFUNDED
#D6216C MAR 5, 2026	 Jacquard-Knit Cotton Jumper QTY: 1 • FENIL PATEL	SIZING ISSUE - TOO LARGE	APPROVED


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- Collections
- Wishlist
- Shopping Cart
- Order History

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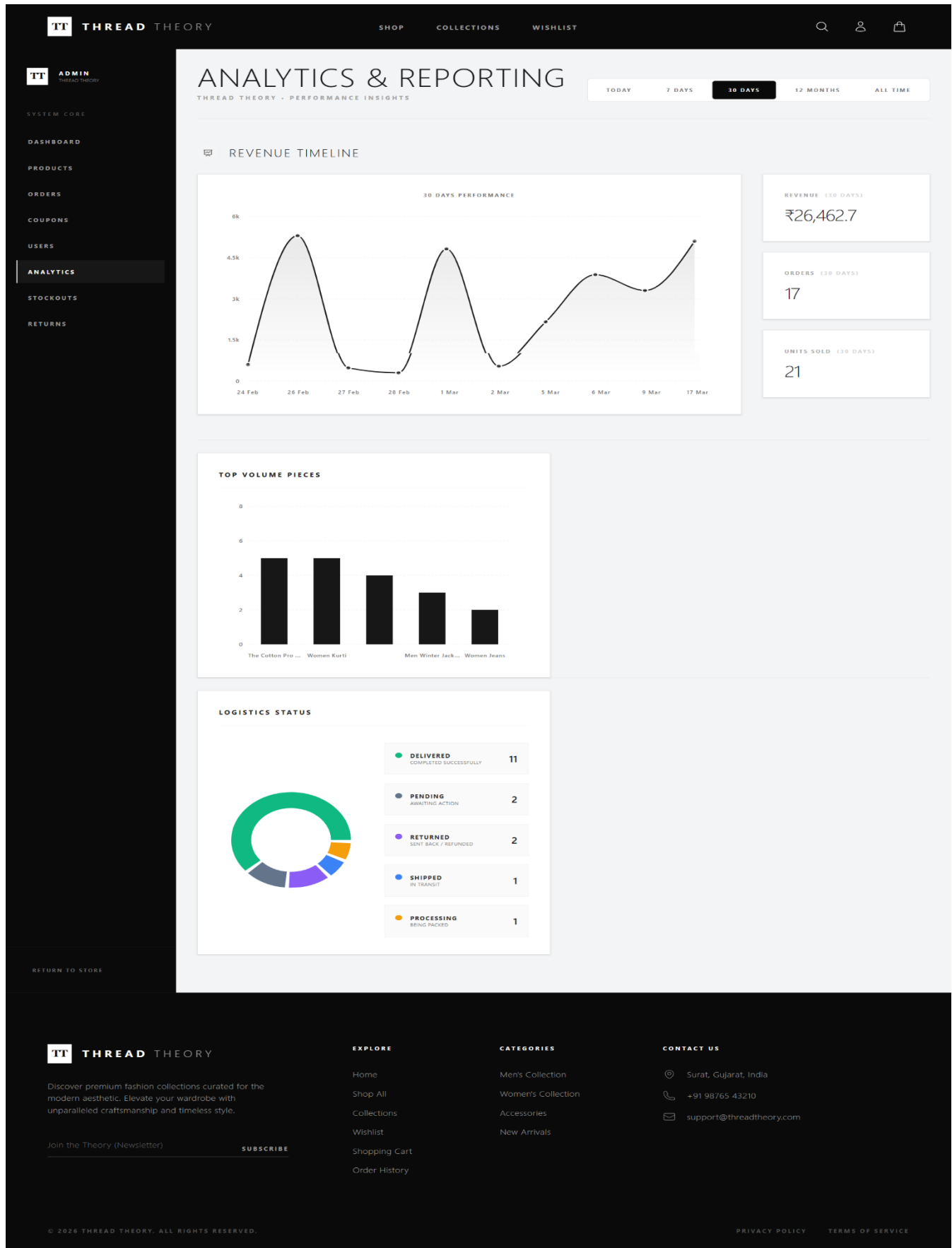
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Analytics Page :



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[SYSTEM CORE](#)

DASHBOARD

[PRODUCTS](#)

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[COUPONS](#)

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[STOCKKOUTS](#)

[RETURNS](#)

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COMMAND CENTER
THREAD THEORY - PERFORMANCE METRICS

TODAY

7 DAYS

30 DAYS

12 MONTHS

ALL TIME

GROSS REVENUE

₹5,097.4

TOTAL ORDERS

2

UNITS SOLD

3

ACTIVE CLIENTS

0

FILTERED TRANSACTIONS

DAILY

ORDER REF	CLIENT	DATE & TIME	AMOUNT
#F46B59	 Fenil Patel	MAR 17, 2026, 10:35 PM	₹3,998.4
#5F3280	 Payal Patel	MAR 17, 2026, 11:18 AM	₹1,099


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[Collections](#)
[Wishlist](#)
[Shopping Cart](#)
[Order History](#)

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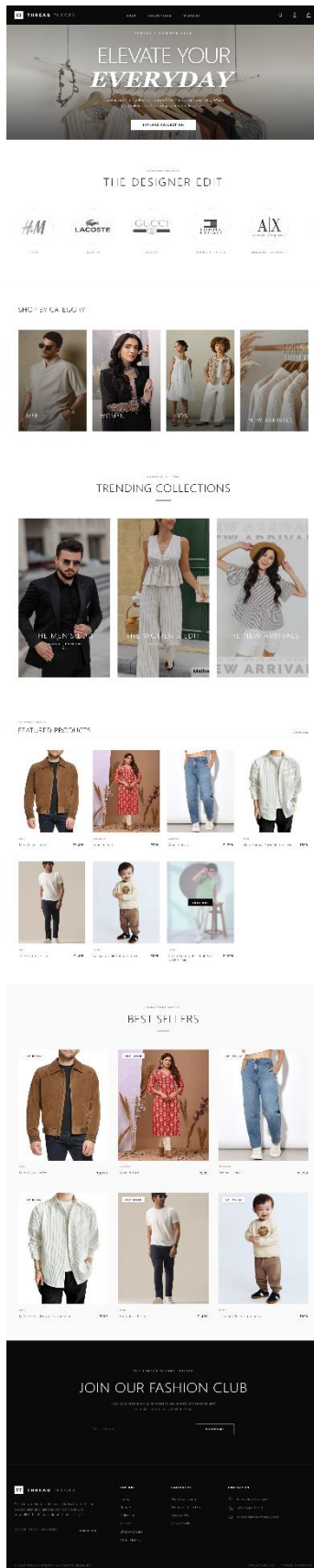
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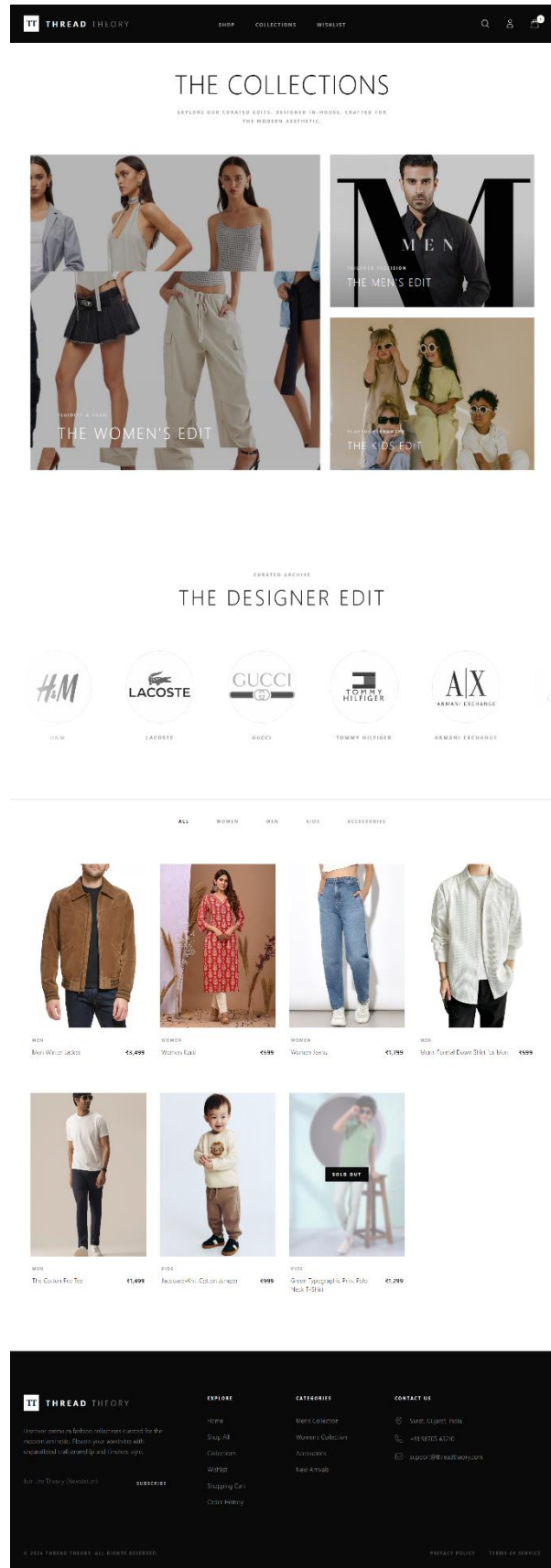
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Client Side

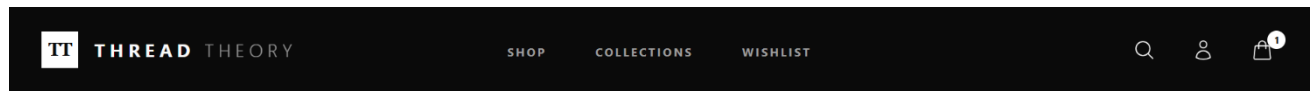
Home Page



Collection page



My Profile Page :



MY PROFILE

WELCOME BACK, MANTHAN



MANTHAN
manthan@gmail.com

EDIT DETAILS

SIGN OUT

ACCOUNT DIRECTORY



ORDER HISTORY

Track your recent purchases and view invoices.



RETURNS & REFUNDS

Track your return requests, view admin notes, and monitor refunds.



SAVED PIECES

View and manage the exclusive items you've curated for later.



SHOPPING BAG

Review your current selections and proceed to secure checkout.



PAYMENT METHODS

Manage your secure payment options and view transaction history.



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Product Info Page :


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Anand is 6'7 wearing size S-ol Regular

THE COTTON PRO TEE

₹1,499

★ 5.0 (1)

The Cotton Pro Tee combines cotton-soft comfort with built-in performance. Its dual-layer fabric feels smooth on the outside, while the inner layer wicks sweat to keep you fresh all day.

SELECT SIZE

S

M

L

XL

2XL

3XL

SIZE GUIDE

ADD TO BAG

 SAVE TO WISHLIST

 VISUAL TRY-ON EXPERIENCE

 COMPLIMENTARY EXPRESS DELIVERY

 14-DAY FREE RETURNS

CLIENT REVIEWS

1 REVIEW • AVERAGE 5.0 / 5

★★★★★

"Very Good Experience I try Visual Try On feature very best That accurate results real wear experience. I'm happy with this product."

JASH

WRITE A REVIEW

RATING

★★★★★

YOUR THOUGHTS

Share details about the fit, quality, and style...

SUBMIT REVIEW

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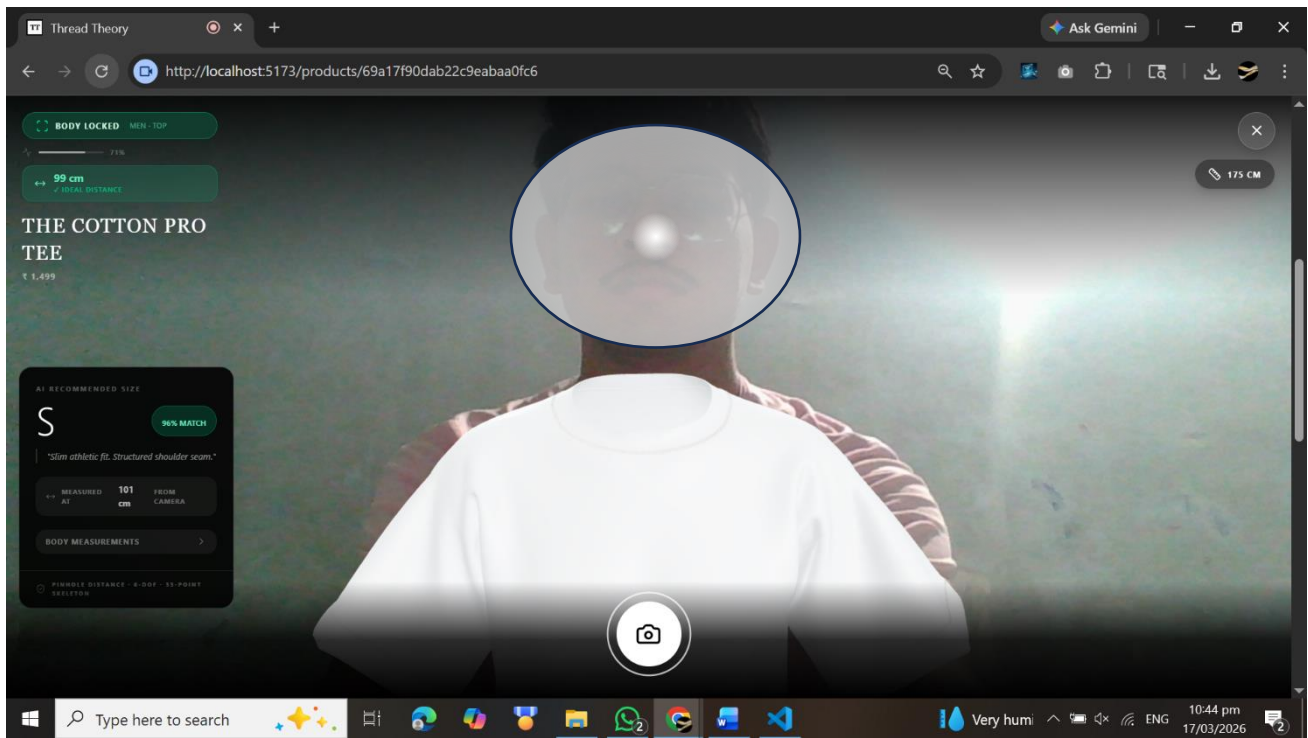
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[Women's Collection](#)
[Accessories](#)
[New Arrivals](#)

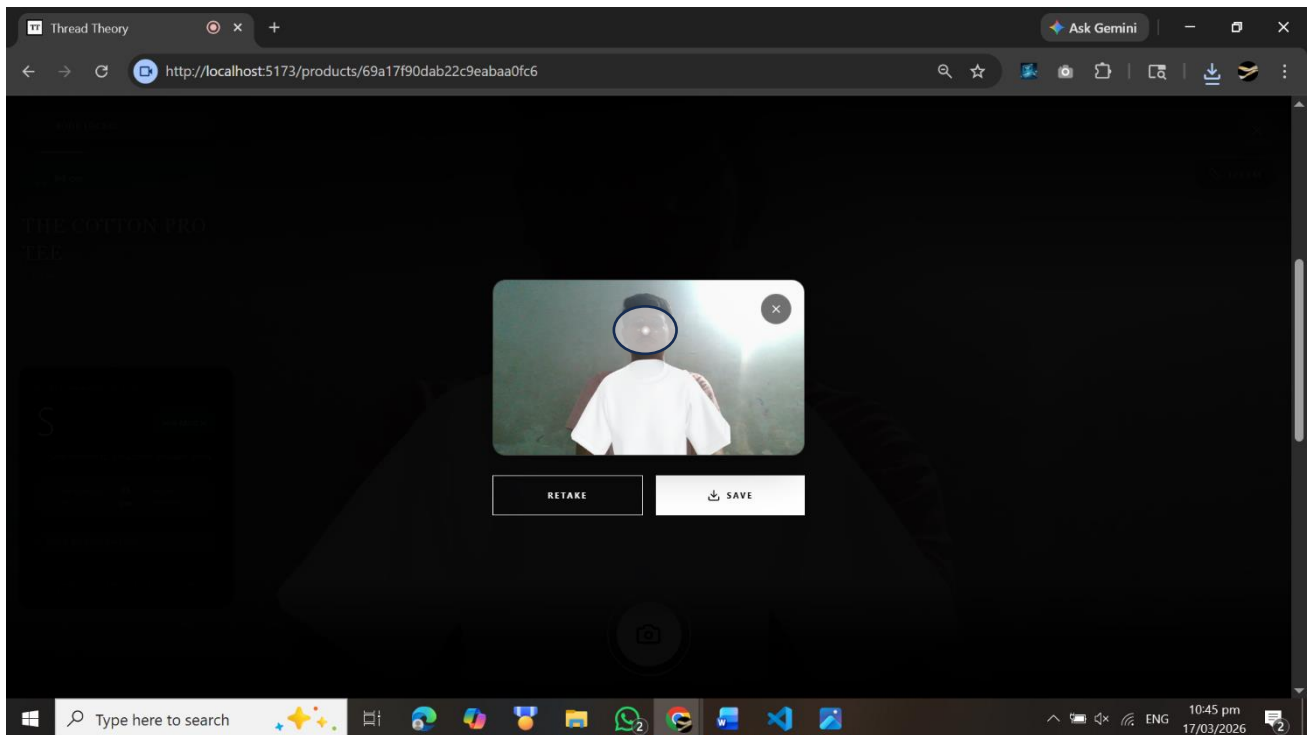
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ORDER HISTORY

YOUR COMPLETE PURCHASING LEDGER

ORDER	DATE	STATUS	TOTAL	ACTION
 #38F5E0	FEBRUARY 28, 2026	● DELIVERED	₹299.5	VIEW DETAILS
 #036C8D	FEBRUARY 27, 2026	● RETURN REQUESTED	₹479.2	VIEW DETAILS TRACK RETURN


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Home

Shop All

Collections

Wishlist

Shopping Cart

Order History

CATEGORIES

Men's Collection

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Accessories

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RETURNS LEDGER

TRACK YOUR DISPUTES AND REFUNDS [VIEW ORDER HISTORY](#)

PRODUCT	REASON	STATUS	UPDATES
 Women Kurti QTY: 1 ORDER #03BCD	Defective or damaged piece MAR 18, 2026	● PENDING	Awaiting review...


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[Home](#)

[Shop All](#)

[Collections](#)

[Wishlist](#)

[Shopping Cart](#)

[Order History](#)

CATEGORIES

[Men's Collection](#)

[Women's Collection](#)

[Accessories](#)

[New Arrivals](#)

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[SHOP](#)
[COLLECTIONS](#)
[WISHLIST](#)





[← BACK TO ORDERS](#)

ORDER #38F5E0

PLACED ON FEBRUARY 28, 2026

PAYMENT
PAID

STATUS
● DELIVERED

ORDER SUMMARY

Reference	2238F5E0
Date	2/28/2026
Total Items	1

DELIVERY ADDRESS

Manthan
 Rajani Plot , Sonal Krupa Travels Office near Jamjodhpur
 Jamjodhpur - 360530

Tel: 9879917331

PURCHASED PIECES



MEN

Men's Formal Down Shirt for Men

QTY: 1

₹599 each

₹599

RETURN ITEM

SUBTOTAL	₹299.5
SHIPPING	COMPLIMENTARY
TOTAL	₹299.5


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EXPLORE

- Home
- Shop All
- Collections
- Wishlist
- Shopping Cart
- Order History

CATEGORIES

- Men's Collection
- Women's Collection
- Accessories
- New Arrivals

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[SHOP](#)
[COLLECTIONS](#)
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SHOPPING BAG

2 ITEMS



MEN
Men Winter Jacket
SIZE: L

—
1
+

₹3,499

REMOVE



MEN
Men's Formal Down Shirt for Men
SIZE: L

—
1
+

₹599

REMOVE

ORDER SUMMARY

Subtotal	₹4,098
Shipping	CALCULATED AT CHECKOUT
Total	₹4,098

[PROCEED TO CHECKOUT](#) →

[OR CONTINUE SHOPPING](#)


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EXPLORE

- Home
- Shop All
- Collections
- Wishlist
- Shopping Cart
- Order History

CATEGORIES

- Men's Collection
- Women's Collection
- Accessories
- New Arrivals

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Secure CheckOut Page:


THREAD THEORY

[SHOP](#)
[COLLECTIONS](#)
[WISHLIST](#)





SECURE CHECKOUT

COMPLETE YOUR PURCHASE IN 3 STEPS

1. DELIVERY DETAILS

[EDIT](#)

Manthan Patel
A/2 Madhav Mandir Soc. , Near Govindji Hall / SMVS BRTS , Dabholi - 395004 , Surat ,Gujarat .,
Surat, 395004
+916354843201

2. SHIPPING METHOD

[EDIT](#)

Standard Delivery (Complimentary)

3. SECURE PAYMENT

Your transaction is encrypted and secure. Click below to open our trusted payment gateway.

PAY ₹3,278.4

ORDER SUMMARY

[EDIT BAG](#)



Men Winter Jacket

SIZE: L | QTY: 1

₹3,499



Men's Formal Down Shirt for Men

SIZE: L | QTY: 1

₹599

WELCOME20

[APPLY](#)

Subtotal	₹4,098
Discount applied	- ₹819.6
Shipping	Complimentary
TOTAL DUE	₹3,278.4


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- Home
- Shop All
- Collections
- Wishlist
- Shopping Cart
- Order History

CATEGORIES

- Men's Collection
- Women's Collection
- Accessories
- New Arrivals

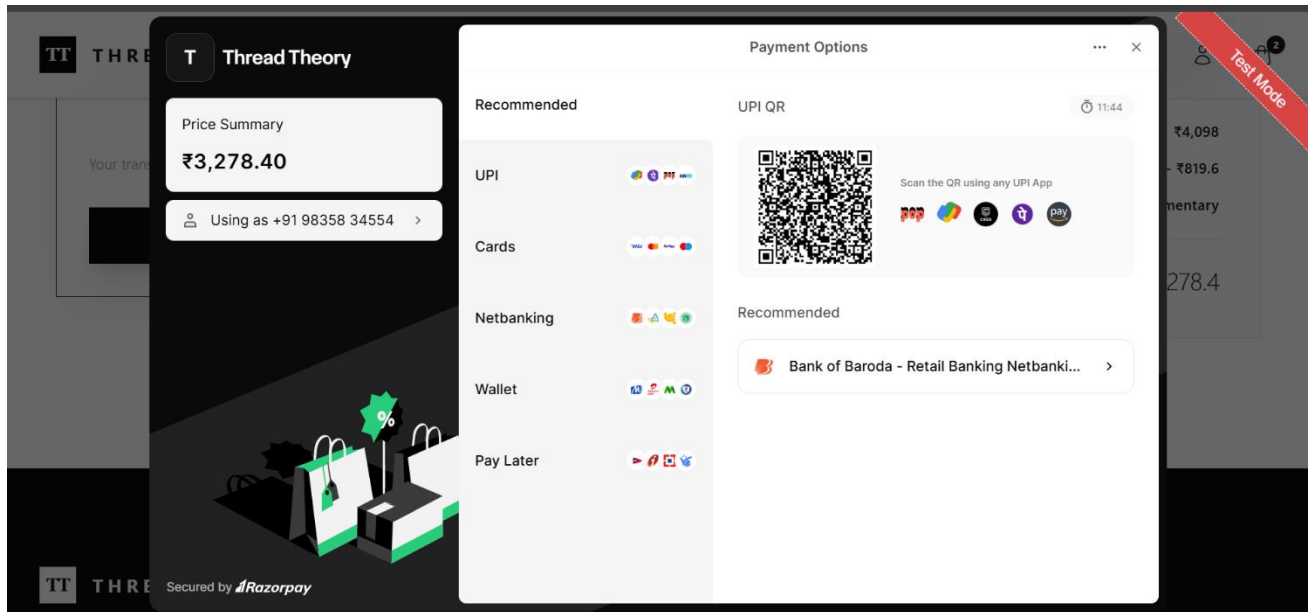
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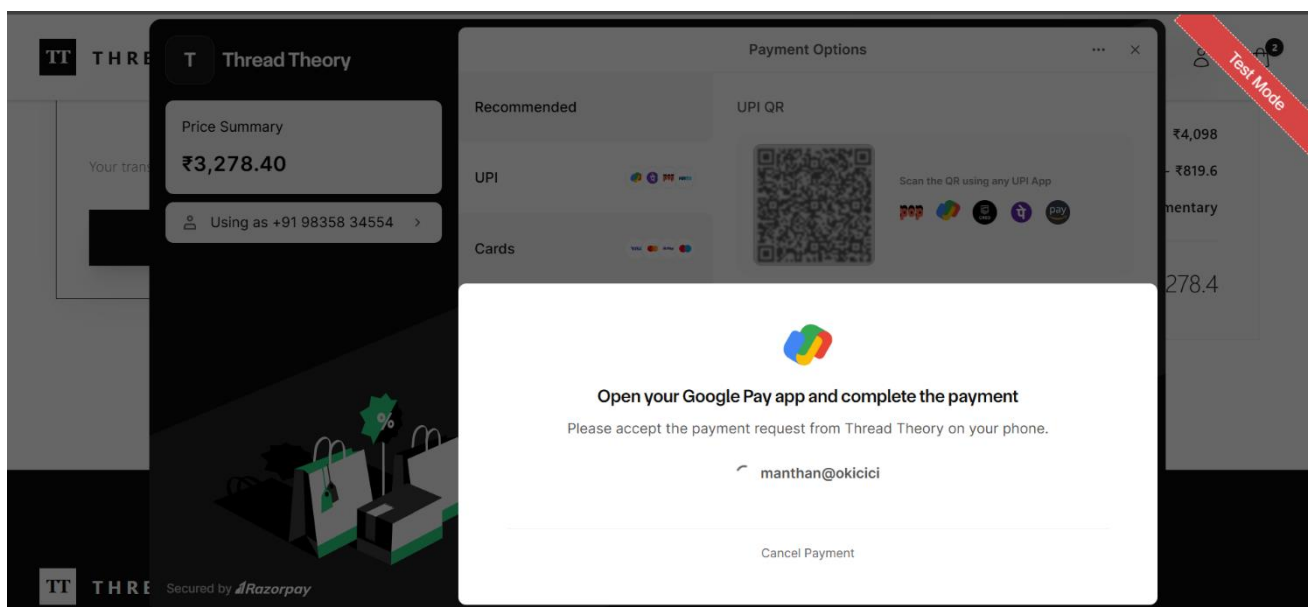
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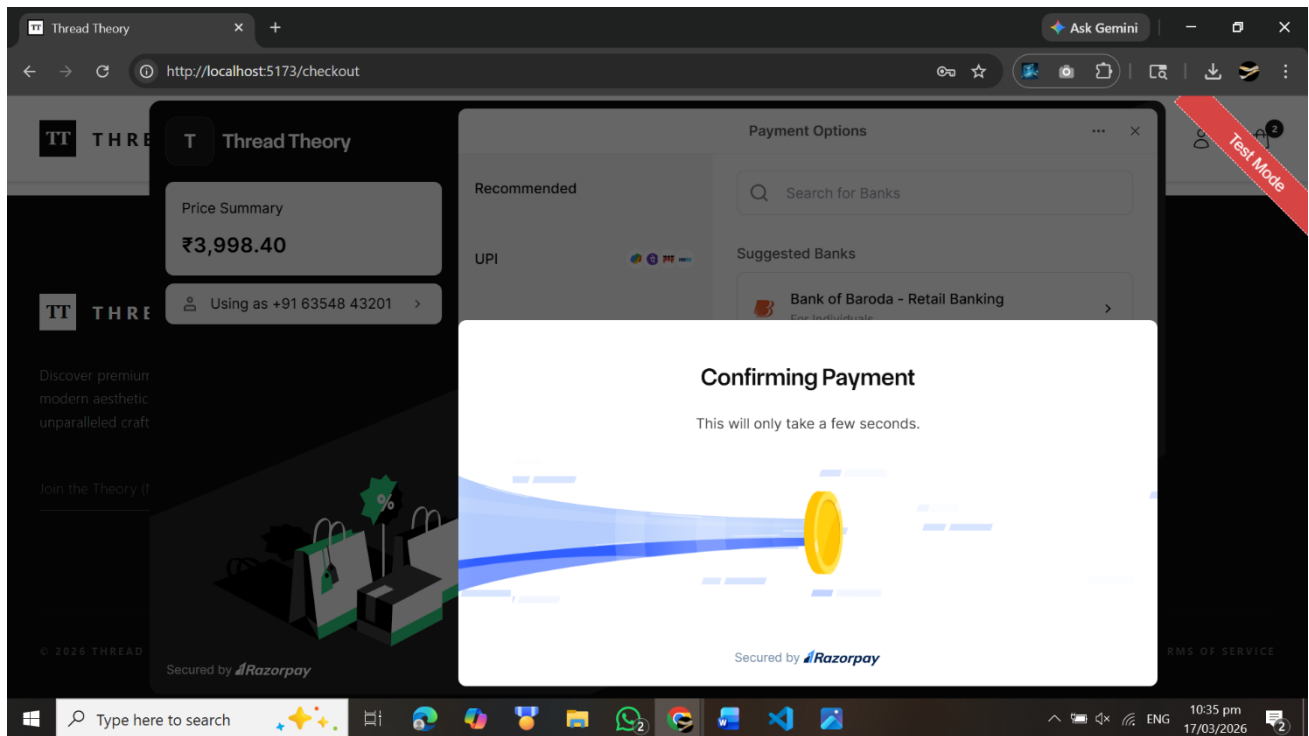
Razorpay Getway Page :



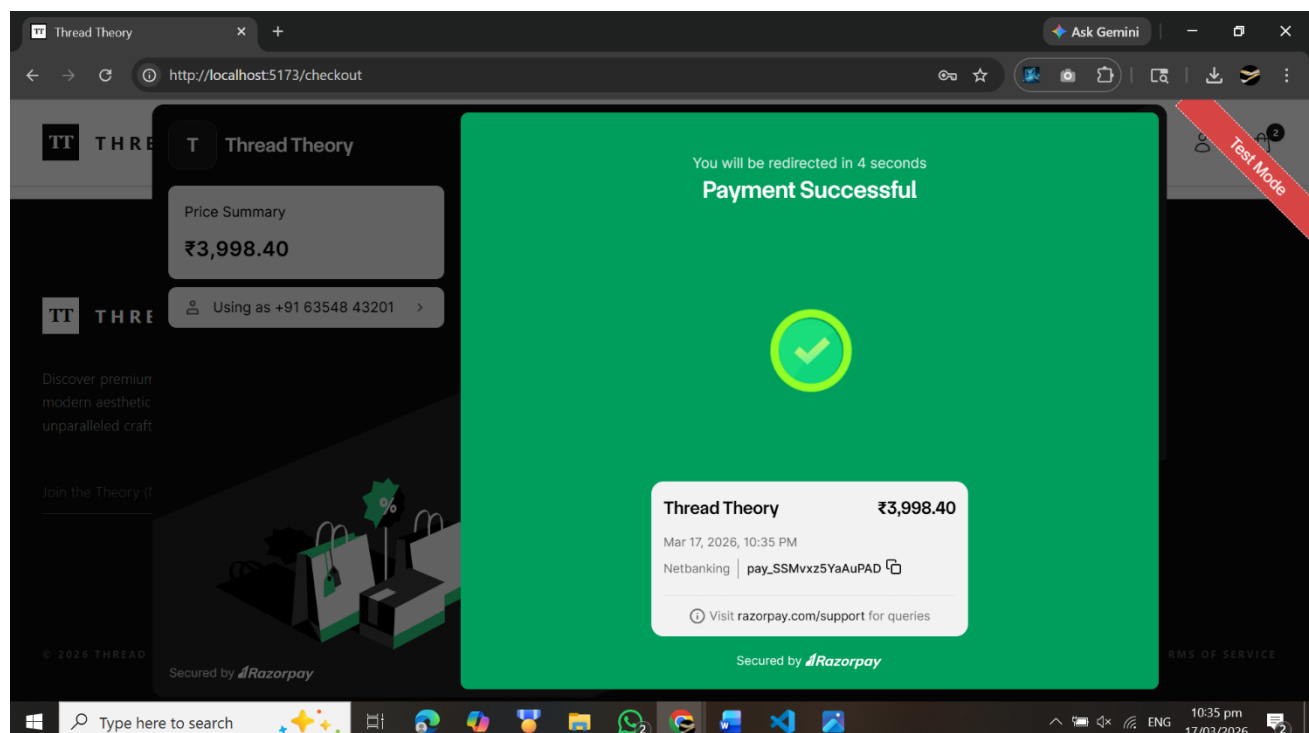
Processing UPI Id :



Payment Process Page :



Payment Pay Confirm :



Order History Page :


THREAD THEORY

[SHOP](#)
[COLLECTIONS](#)
[WISHLIST](#)





[HOME](#) / [ACCOUNT](#) / **ORDER HISTORY**

ORDER HISTORY

YOUR COMPLETE PURCHASING LEDGER

ORDER	DATE	STATUS	TOTAL	ACTION
 #9D5E29	MARCH 25, 2026	● PENDING	₹3,278.4	VIEW DETAILS
 #38F5E0	FEBRUARY 28, 2026	● DELIVERED	₹299.5	VIEW DETAILS
 #036C8D	FEBRUARY 27, 2026	● RETURN REQUESTED	₹479.2	VIEW DETAILS TRACK RETURN


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- Home
- Shop All
- Collections
- Wishlist
- Shopping Cart
- Order History

CATEGORIES

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- Women's Collection
- Accessories
- New Arrivals

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[SHOP](#)
[COLLECTIONS](#)
[WISHLIST](#)





[HOME](#) / [ACCOUNT](#) / [PAYMENT LEDGER](#)

PAYMENT LEDGER

YOUR SECURE TRANSACTION HISTORY [VIEW ORDER HISTORY](#)

ORDER REF	DATE	STATUS	AMOUNT	ACTION
#9D5E29	MARCH 25, 2026	● PAID	₹3,278.4	VIEW ORDER
#38F5E0	FEBRUARY 28, 2026	● PAID	₹299.5	VIEW ORDER
#036C8D	FEBRUARY 27, 2026	● PAID	₹479.2	VIEW ORDER


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[Shop All](#)

[Collections](#)

[Wishlist](#)

[Shopping Cart](#)

[Order History](#)

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[Women's Collection](#)

[Accessories](#)

[New Arrivals](#)

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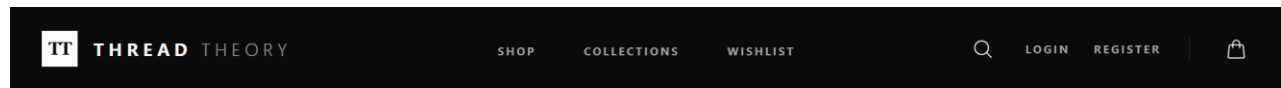
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Other Support Page

Privacy Policy :



PRIVACY POLICY

THREAD THEORY • DATA PROTECTION & SECURITY

Last Updated: October 2026

CONTENTS

- 01. OUR COMMITMENT
- 02. INFORMATION WE COLLECT
- 03. HOW WE USE YOUR DATA
- 04. INFORMATION SHARING
- 05. COOKIES & TRACKING
- 06. DATA SECURITY
- 07. YOUR RIGHTS & CHOICES

01. OUR COMMITMENT

At Thread Theory, we respect your privacy and are committed to protecting the personal information you share with us. This Privacy Policy outlines our practices regarding the collection, use, and safeguarding of your data when you interact with our Site, purchase our garments, or communicate with our concierge team.

02. INFORMATION WE COLLECT

We collect information to provide a more personalized and seamless luxury shopping experience. This includes:

- **Personal Details:** Your name, email address, phone number, shipping and billing addresses provided during checkout or account creation.
- **Transaction Data:** Details about the pieces you purchase, sizing preferences, and payment status (processed securely via Razorpay; we do not store your full card details).
- **Automated Data:** IP addresses, browser types, device identifiers, and browsing behavior collected via cookies to enhance site functionality.

03. HOW WE USE YOUR DATA

The data we collect is utilized exclusively to elevate your experience with Thread Theory. Specifically, we use your information to:

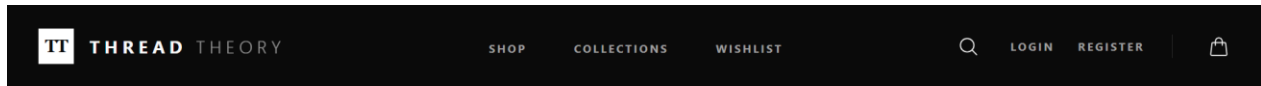
- Process and fulfill your orders, including sending shipping updates and delivery confirmations.
- Manage your returns, exchanges, and customer service inquiries efficiently.
- Tailor our editorial content and product recommendations to your sartorial preferences.
- Send you exclusive invitations, collection previews, and promotional materials (only if you have opted in).

04. INFORMATION SHARING

Thread Theory does not sell, rent, or trade your personal information to third parties. We only share your data with trusted partners who assist us in operating our platform and conducting our business, such as payment gateways (Razorpay), premium courier services, and secure cloud hosting providers. These partners are strictly obligated to keep your information confidential and secure.

05. COOKIES & TRACKING

Terms & Conditions :



TERMS & CONDITIONS

THREAD THEORY • LEGAL FRAMEWORK & POLICIES

Last Updated: October 2026

CONTENTS

- 01. ACCEPTANCE OF TERMS
- 02. INTELLECTUAL PROPERTY
- 03. PRODUCTS & PRICING
- 04. ORDERS & PAYMENTS
- 05. SHIPPING & DELIVERY
- 06. RETURNS & EXCHANGES
- 07. USER CONDUCT
- 08. LIMITATION OF LIABILITY

01. ACCEPTANCE OF TERMS

By accessing, browsing, or using the Thread Theory website (the "Site"), you acknowledge that you have read, understood, and agree to be bound by these Terms and Conditions. If you do not agree to these terms, please do not use our Site.

We reserve the right to update or modify these Terms at any time without prior notice. Your continued use of the Site following any changes constitutes your acceptance of the revised terms.

02. INTELLECTUAL PROPERTY

All content on this Site, including but not limited to text, graphics, logos, images, audio clips, digital downloads, data compilations, and garment designs, is the exclusive property of Thread Theory and is protected by international copyright and trademark laws. You may not reproduce, duplicate, copy, sell, or otherwise exploit any portion of this Site without express written consent from us.

03. PRODUCTS & PRICING

We strive to display the colors, textures, and details of our garments as accurately as possible. However, we cannot guarantee that your monitor's display of any color will be perfectly accurate.

All prices are shown in INR (₹) and are inclusive of applicable taxes unless otherwise stated. We reserve the right to change the prices of our products at any time without notice. In the event of a pricing error on the Site, Thread Theory reserves the right to cancel any orders placed for the item at the incorrect price.

04. ORDERS & PAYMENTS

Placing an order on our Site constitutes an offer to purchase. We reserve the right to refuse or cancel any order for any reason, including limitations on quantities available for purchase, inaccuracies, or errors in product or pricing information, or problems identified by our fraud avoidance department.

We utilize Razorpay as our secure payment gateway. By submitting your payment information, you grant us the right to provide such information to third parties for purposes of facilitating the completion of your purchase.

05. SHIPPING & DELIVERY

Thread Theory offers complimentary shipping on selected orders. Delivery timelines are estimates and commence from the date of shipping, rather than the date of order. We are not responsible for delays caused by customs clearance processes or courier delays. Risk of loss and title for items purchased from our Site pass to you upon delivery of the items to the carrier.

Concierge :

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SHOP

COLLECTIONS

WISHLIST



LOGIN

REGISTER



CONCIERGE

THREAD THEORY • CLIENT SERVICES

Whether you require assistance with an existing order, sizing advice, or wish to inquire about our latest collections, our dedicated concierge team is at your complete disposal.

DESIGN STUDIO

Thread Theory Headquarters
VIP Road, Vesu
Surat, Gujarat 395007
India

EMAIL INQUIRIES

concierge@threadtheory.com

TELEPHONE

+91 98765 43210

SEND A MESSAGE

FULL NAME *

EMAIL ADDRESS *

ORDER REFERENCE (OPTIONAL)

YOUR MESSAGE *

SEND INQUIRY**TT** THREAD THEORY

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Software Testing

Software testing for **Thread Theory** is a critical phase designed to ensure that the platform delivers a seamless, secure, and luxury experience for both clients and administrators. The testing strategy verifies that all e-commerce functionalities—from real-time product searching to secure checkout and inventory management—operate flawlessly under various conditions.

7.1 Unit Testing

Unit testing involves verifying the smallest testable parts of the application in isolation to ensure they function exactly as intended. For Thread Theory, this focuses on individual React components, custom hooks, and backend utility functions.

- **Frontend Validation Logic:** Testing the strict regex patterns in the `validateForm()` functions (e.g., ensuring passwords require 8 characters, symbols, and numbers, and that phone numbers only accept 10 digits).
- **Cart Quantity Constraints:** Verifying that the `handleQtyChange` function in the Shopping Bag strictly prevents users from selecting fewer than 1 item or exceeding the maximum limit of 10 items per order.
- **Backend Utility Functions:** Testing the `getDateFilter()` function in the Admin Controller to ensure it accurately calculates date ranges (e.g., exactly 7 days or 30 days) for the dashboard analytics.
- **State Management:** Ensuring the `useCart` and `useAuth` context providers accurately initialize, update, and clear states upon user login or logout.

7.2 Integration Testing

Integration testing evaluates how different modules and services within the MERN stack interact with one another. This ensures that data flows correctly between the MongoDB database, the Express/Node.js backend API, and the React frontend.

- **Authentication & Authorization Flow:** Testing the login process to ensure that when a user is marked as `isBlocked: true` in the database, the backend successfully throws a 403 Forbidden error, and the frontend accurately displays the "Account Suspended" toast notification.

- **AJAX Real-Time Search:** Verifying that the debounced search input in the Navbar correctly fetches and filters the product list from the /products API endpoint without causing excessive server requests or UI lag.
- **Inventory Synchronization:** Ensuring that when an Admin updates stock via the "Restock" modal in AdminStockOut.jsx, the database reflects the new value and the "Out of Stock" warning instantly disappears from the client-facing product page.
- **Return Management Interaction:** Testing the flow where an Admin approves a return, ensuring the custom admin note is successfully saved to the database and the order status updates uniformly across all related tables.

7.3 System Testing

System testing is the end-to-end evaluation of the fully integrated Thread Theory application to verify that it meets all technical and business requirements.

- **End-to-End (E2E) Checkout Journey:** Simulating a complete user journey: A client logs in -> searches for a piece -> adds it to the bag -> proceeds to checkout -> completes payment -> and the Admin successfully views the transaction in the Command Center.
- **UI/UX & Responsiveness Testing:** Rigorously testing the frontend across various devices and screen zooms (specifically 100% zoom on 1366x768 screens) to ensure the fluid tables, slide-in mobile drawers, and central modals never break layout or require awkward scrolling.
- **Security & Session Management:** Verifying that JSON Web Tokens (JWT) are securely passed in headers, expired tokens automatically log the user out, and non-admins are strictly blocked from accessing /admin/* routes.
- **Performance & Load Testing:** Ensuring the database aggregation pipelines (used for calculating Admin Dashboard revenue and dynamic sales trends) load quickly even with thousands of mock order records.

Future Scope of Enhancements

While Thread Theory currently boasts a robust, high-performance architecture, e-commerce is constantly evolving. The platform is designed with scalability in mind, allowing for the following future integrations:

- **AR Virtual Try-On:** Utilizing the existing `model3Durl` field in the Product database schema to integrate WebGL/Three.js, allowing clients to view 3D models of clothing and accessories in their physical space via their mobile cameras.
- **AI-Powered Style Concierge:** Implementing machine learning algorithms to analyze a client's past purchase history and Wishlist data to provide hyper-personalized outfit recommendations.
- **Advanced Payment Infrastructure:** Upgrading the payment gateway to support multi-currency checkouts for international shipping, as well as integrating "Buy Now, Pay Later" (BNPL) options or EMI features.
- **Automated Logistics Tracking:** Integrating third-party shipping APIs (like Shiprocket or FedEx) to provide real-time, step-by-step SMS and email tracking notifications for dispatched orders.
- **Progressive Web App (PWA) Capabilities:** Enhancing the frontend with service workers and offline caching so clients can browse the catalog and view their cart even with unstable internet connections.

Conclusion

The development of **Thread Theory** successfully bridges the gap between luxury fashion retail and cutting-edge web technology. By leveraging the robust capabilities of the MERN stack (MongoDB, Express.js, React.js, and Node.js), this project has culminated in a highly responsive, secure, and scalable e-commerce platform designed to deliver a premium user experience.

Throughout the project lifecycle, equal priority was given to both the client-facing storefront and the administrative backend. For the clients, the platform offers a seamless shopping journey characterized by intuitive navigation, real-time AJAX product searching, secure JWT-based authentication, and a frictionless checkout process. The minimalist, high-contrast UI design ensures that the focus remains entirely on the clothing collections, mirroring the aesthetic of top-tier fashion brands.

From an operational standpoint, the **Admin Command Center** provides Thread Theory's management team with complete control over the business. The implementation of dynamic sales analytics, automated depleted stock alerts, and a streamlined dispute-management system for returns significantly reduces administrative overhead. Furthermore, by building custom, strict validation logic and bypassing standard browser defaults, the platform guarantees data integrity and security across all user inputs.

Ultimately, Thread Theory proves to be much more than a standard digital storefront. It is a comprehensive, full-stack e-commerce ecosystem. With its highly organized database architecture and modular React frontend, the platform is not only fully equipped to handle current retail operations but is also perfectly positioned to scale and integrate future enhancements as the brand grows.

Reference

- <https://reactnative.dev/docs/>
- <https://nodejs.org/en/docs/>
- <https://expressjs.com/>
- <https://www.mongodb.com/docs/manual/>
- <https://tailwindcss.com/docs>
- <https://developer.mozilla.org/>
- <https://jwt.io/>
- <https://www.npmjs.com/package/bcryptjs>